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Common Institutional Ownership and Product Market Threats

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Abstract. The common ownership of firms can have anticompetitive effects by incentivizing collusive outcomes that maximize joint surpluses of the commonly held firms or procompetitive effects through enhanced knowledge spillovers. Using a difference-in-differences regression methodology that exploits mergers between financial institutions as exogenous shocks to common ownership, our baseline results suggest that higher common ownership leads to greater product market fluidity (a text-based metric of competition) and generally leads to more product development and higher investments. These findings suggest that, on average, common ownership spurs dynamism in product spaces rather than tacit collusion between cross-held competitors. This is especially true in economic environments in which it is easier to take advantage of knowledge spillovers. However, common ownership can also inhibit product market competition and dynamism, especially in industries more prone to quasi-monopoly outcomes in product spaces. Implementing a one-size-fits-all regulatory policy limiting common ownership may be harmful in industries with strong spillover opportunities.

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1. Introduction

The secular rise of delegated portfolio management in the United States has resulted in large institutional investors holding significant stakes in competing firms. The pervasiveness of these institutional investors ("common owners") challenges the classic notion about corporations' objective function of maximizing a single firm's value (Berle and Means 1932) because institutional investors naturally want to maximize the combined value of their holdings in all firms in their portfolios. Since a firm's strategic decisions impact not only the value of the firm, but also the value of its competitors, common owners may pursue anticompetitive strategies to achieve quasi-monopoly rents in product markets (market power hypothesis).[1]

This view, however, neglects the welfare-promoting externalities stemming from institutions' shareholdings in competing firms (Vives 2020). Individual firms, acting in their own interest, may refrain from making costly investments for fear of knowledge spillovers to rivals (Bloom et al. 2013). Common owners, by internalizing these externalities, can potentially promote investments in technologies with strong spillover effects for all firms in their portfolios (López and Vives 2019). Additionally, they can facilitate product market collaborations and information-sharing between firms in their portfolio and, thus, increase innovation productivity (He and Huang 2017). Consequently, institutional cross-holdings of public firms can also have an efficiency enhancing effect (efficiency hypothesis). A comprehensive examination of common owners' product market implications, therefore, requires the joint analysis of both the market power and efficiency hypotheses. In this paper, we empirically examine the market power and efficiency channels in a broad cross-section of publicly traded firms in which we can assess common ownership's net product market effects as well as its heterogeneous effects along multiple dimensions of industry characteristics.

Extant empirical studies largely analyze whether firms with higher common ownership possess market power to set higher output prices and/or generate greater price–cost margins (markup). However, under the efficiency channel, common owners can promote greater investments and knowledge spillovers between competitors and, thus, reduce production costs and increase markup. Thus, a positive relation between common ownership and price–cost margins can represent support for either hypothesis. As such, a holistic view of the competitive effects of common ownership can be achieved by studying strategic variables beyond price and quantity for firms in distinctive economic environments (Vives 2020).

We, therefore, analyze the product market implications of common ownership along a new dimension: the competition and dynamism in product spaces. To capture rich competitive dynamics between firms for a broad set of industries, we primarily use the product market fluidity variable (*fluidity*) developed by Hoberg et al. (2014) as a measure of competitive risk or threats faced by a firm. Fluidity measures the extent to which a firm's competitors are changing their product description in annual reports (10-Ks) relative to the firm's own product description. A firm faces greater product market threats when its competitors' actions (reflected in the changes in their product description) significantly disrupt the firm's current product offerings (reflected in the firm's own product description). Thus, higher values for fluidity are associated with greater "movements" or "vibrations" by competitors in a firm's product spaces. Fluidity allows us to capture product market threats for almost all public firms based on their required annual 10-K reports, thereby letting us examine the heterogeneous effects of common ownership along various dimensions of industry characteristics.[2]

In our empirical predictions, the net influence of an increase in common ownership on fluidity is determined by its effect on two countervailing forces: higher spillover opportunities and enhanced market power incentives. On the one hand, higher investments and greater information sharing because of enhanced spillover opportunities that arise from an increase in overlapping ownership create a "rat race" among competitors in the same economic sector, pushing firms to disrupt each other's product market spaces more actively. The increased disruption can occur through product innovation in existing product spaces and/or the creation of new products and markets. Specifically, in a dynamic process, the actions of rivals hasten the process of product innovation because the focal firm must introduce new and/or improved products if it wants to grow and vice versa (Aghion and Howitt 1992, Argente et al. 2024) and, consequently, lead to greater output, product innovation, and welfare (Leahy and Neary 1997). The higher spillover opportunities resulting from an increase in common ownership further spur this dynamic process and lead to a more vibrant product market. The efficiency channel, thus, predicts that an increase in common ownership will generate greater product market competition as evidenced by higher fluidity.

On the other hand, the market power channel implies that increased common ownership enhances anticompetitive motives through the formation of a stronger collusive alliance between firms in their portfolios, thereby allowing these firms to earn higher profits. This increased market power can manifest itself in two distinct ways. First, as classic industrial organization theories (e.g., Hotelling 1929, Chamberlin 1933, Tirole 1988) suggest, firms keep their products differentiated to steer away from price competition, thereby allowing them to create a monopolistic market in each product category. Thus, an increase in common ownership strengthens market power incentives, and consequently, portfolio firms compete less with each other in product spaces by further differentiating their products, thereby leading to greater fluidity (product differentiation channel). Notably, the increase in fluidity in this context is also a manifestation of competitive pressures but with the intent to reduce and not increase competition. Second, economists have long maintained that monopolists have greater incentives to maintain the status quo to protect their financial interests; as Hicks (1935, p. 8) comments, "The best of all monopoly profits is a quiet life." Accordingly, an increase in common ownership results in a decrease in fluidity because of decreased incentives of commonly owned firms to compete (quiet life channel).

Our baseline predictions suggest that an increase in common ownership will result in higher fluidity under both the efficiency and product differentiation channels. Under such a scenario, we need to use another construct—product similarity—developed by Hoberg and Phillips (2010) to determine which of these two effects is the dominant channel in our baseline tests. They compute product similarity between any two firms as the cosine similarity between the word vectors of two firms. The average product similarity of a focal firm with its rivals is then computed as the average of the product similarity between the firm and each of its rivals. The product differentiation channel explicitly predicts that an increase in common ownership should result in a decrease in average similarity. The efficiency channel, however, is consistent with either an increase or decrease in average similarity because it only has predictions regarding increased movements by product market rivals in response to the focal firm's product offerings.

We empirically examine the relation between common ownership and competitive threats faced by a firm by using a sample of U.S. public firms over the period 1997 to 2017. Our main empirical tests exploit multiple merger events between financial institutions as exogenous shocks that can increase common ownership (see, e.g., He and Huang 2017, Lewellen and Lowry 2021). Specifically, we estimate difference-in-differences (DID) regression models to both alleviate selection issues and control for unobservable factors that may simultaneously affect the institutional cross-holdings of competitors and the product market threats from competitors. Lewellen and Lowry (2021) argue that these mergers provide the most robust identification strategy in common ownership research.

Following an approach similar to that described in He and Huang (2017) and Lewellen and Lowry (2021), we identify a treatment group of firms that experience an increase in common ownership with their competitors because of institutional mergers in our DID models. We use three measures of common ownership: the modified Herfindahl–Hirschman index delta (*MHHID*) of O'Brien and Salop (2000), the cross-ownership (*Cross*) measure of Hansen and Lott (1996), and a recent measure of common ownership with managerial incentives (*GGL*) developed by Gilje et al. (2020) to validate our experimental setting. We document a significant increase in all three measures of common ownership for the treatment group after institutional mergers, thereby suggesting that these mergers represent positive shocks to common ownership.

More importantly, we find that the treatment group experiences higher fluidity following institutional mergers. This result suggests a causal and net positive impact of an exogenous increase in common ownership on fluidity. This net positive impact is consistent with both the efficiency and product differentiation channels but inconsistent with the quiet life channel. Additionally, we find that the treatment group experiences a significant increase in average similarity. This latter result is inconsistent with the product differentiation channel. Thus, our baseline results suggest that exogenous shocks to common ownership increase product market competition, which is consistent with the efficiency channel as the dominant motive.

Since the efficiency and market power channels are not mutually exclusive; that is, they can be in play at the same time, we further analyze how the net competitive effects of increases in common ownership vary along dimensions related to common owners' incentives to take advantage of spillovers and their ability to reap monopoly rents. We estimate triple-difference DID regression models to analyze how industry attributes impact the relation between common ownership and fluidity. We find that the positive impact of common ownership on product market threats documented earlier is stronger in industries more conducive to spillover effects, specifically, in less concentrated industries and industries with greater technological proximity between competitors. We also find that the positive impact of common ownership on fluidity becomes weaker in industries in which the market power channel of common ownership is likely to be salient; that is, in industries with either high industry concentration, entry costs, or consumer switching costs. These findings are consistent with both the efficiency and quiet life channels but inconsistent with the product differentiation channel. Overall, the anticompetitive effect of common ownership warrants close investigation only when industry structure facilitates collusive behaviors.

Finally, we examine the impact of common ownership on both new product development and firm investments. Specifically, we trace the evolution of new product development via the size (in terms of number of words) of firms' 10-K product descriptions (Hoberg and Phillips 2010). We find a weak positive impact of a positive exogenous common ownership shock on new product development. Our cross-sectional tests show that a positive common ownership shock enhances (diminishes) new product development when firms have greater (lesser) potential for industry spillover effects. Further, we generally find that the positive common ownership shock has a positive impact on investments. This effect on investments increases (decreases) in less (more) concentrated industries and in industries with higher (lower) technology proximity. These results provide strong corroboration for the conclusions we reach about the product market consequences of common ownership with fluidity as the key strategic variable.

This paper makes the following contributions. We directly contribute to the recent literature on the competitive effects of common ownership. Azar et al. (2018, 2022a) show that common ownership concentration increases product prices in the airline and banking industries. However, other papers provide mixed evidence about the competitive effect of common ownership (e.g., Gramlich and Grundl 2017, Kennedy et al. 2017, Aslan 2019, Koch et al. 2021, Lewellen and Lowry 2021, Dennis et al. 2022). We add to this literature by showing that, on average, common ownership induces more dynamic product spaces. As such, our paper complements He and Huang (2017), who focus on potential information spillovers and find evidence consistent with welfare-enhancing effects of common ownership.

Various global regulatory agencies, including the Federal Trade Commission and the Organization for Economic Co-operation and Development, have held discussions regarding the need for placing restrictions on common ownership beyond the provisions in existing merger guidelines. Our paper suggests that caution needs to be exercised in proposing and implementing any additional regulatory requirements on the ownership of competing firms by financial institutions because they can potentially be welfare-destroying in industries with strong spillover opportunities. In addition, we contribute to some seminal studies (e.g., Hotelling 1929, Chamberlin 1933, Hicks 1935) that suggest firms can exert market power through either a quiet life or product differentiation. In the context of common ownership concentration, we find support for the quiet life channel but not the product differentiation channel.

This paper also fits into the broader literature on the influence of common institutional ownership on corporate decisions (e.g., Matvos and Ostrovsky 2008; Harford et al. 2011; Antón and Polk 2014; He et al. 2019; Kostovetsky and Manconi 2020; Freeman 2021; Antón et al. 2022, 2023). By focusing on product market threats faced by firms rather than their prices and output, we can examine the impact of common ownership on industry competition for a broad cross-section of industries, thereby allowing us to provide new insights on the economic conditions under which higher common ownership can either enhance or reduce competition. In the process, we provide evidence consistent with some of the predictions in the theoretical model of López and Vives (2019) about the impact of common ownership on firm investments. Finally, our paper adds to the literature on common ownership-like effects that arise mainly through common venture capitalists in a private firm setting (see, e.g., Lindsey 2008, Bubna et al. 2020, Kamepalli et al. 2022, Li et al. 2023).

2. Hypotheses Development

2.1. Common Ownership and Fluidity: Baseline Case

The efficiency hypothesis suggests that higher levels of common ownership improve knowledge spillovers and promote investments with positive externalities. Common owners can also facilitate information flows between firms in their portfolios and foster interfirm collaboration such as strategic alliances (He and Huang 2017). Common owners may facilitate managerial learning from peers, thereby making rivals' actions in product markets more responsive to the focal firm's products. In addition, common owners can promote investment with positive spillover effects even without direct communication between common owners and managers. By internalizing the externality from rivals through reducing the adverse impact of involuntary knowledge spillovers to rivals, common ownership can provide more incentives for managers to invest in cost-cutting technologies (López and Vives 2019).

The higher investments and greater information sharing because of enhanced spillover opportunities that arise from an increase in overlapping ownership make firms more responsive to changes in their competitive landscape and push them to disrupt each other's product market spaces more actively. The increased disruption can take place through product innovation in existing product spaces and/or the creation of new products/markets. In general, the actions of rivals quicken the process of product innovation because the focal firm must introduce new and/or improved products if it wants to grow and vice versa (Aghion and Howitt 1992, Argente et al. 2024), leading to greater output, product innovation, and welfare (Leahy and Neary 1997). The higher spillover opportunities resulting from an increase in common ownership further spur this process and lead to a more vibrant product market. We use the fluidity variable developed by Hoberg et al. (2014) as a measure of competitive threats. The efficiency channel, thus, predicts that an exogenous increase in common ownership will generate greater product market competition as captured by higher fluidity.

Hypothesis 1A. *Under the efficiency channel, an increase in common ownership between a firm and its product market competitors has a positive impact on fluidity.*

The market power hypothesis predicts that common owners can incentivize managers to soften competition with industry rivals. By forming an implicit collusive alliance between firms in their portfolios, common owners can mitigate product market threats to earn high price–cost margins and profits (see, e.g., Schmalz 2018, 2021 for an extensive survey of the literature). Despite setting prices and quantities as a monopolist, firms with high common ownership can also keep their products differentiated to steer away from competition in the same product space, thereby allowing them to create a monopolistic market (e.g., Hotelling 1929, Chamberlin 1933, Tirole 1988). We call this the product differentiation channel. Additionally, economists have long asserted that monopolists have greater incentives to maintain the status quo to protect their financial interests; as Hicks (1935, p. 8) comments, "The best of all monopoly profits is a quiet life." We call this the quiet life channel. In sum, both these market power–related arguments suggest that a positive exogenous shock to common ownership enhances market power incentives and, consequently, results in less competitive product markets.

However, the impact of the common ownership shock on fluidity is exactly the opposite under these two market power channels. Higher fluidity can result from the commonly held rival firms either getting closer to or further from the focal firm's product spaces; that is, there is more movement in its product market. Reduced competition under the quiet life channel suggests that increased common ownership results in rivals exerting increased market power through less movement in the product market, thereby resulting in a decrease in fluidity.

Hypothesis 1B. *Under the quiet life channel, an increase in common ownership between a firm and its product market competitors has a negative effect on fluidity.*

In contrast, under the product differentiation channel, rivals exert greater market power arising from increased common ownership by further differentiating their products to reduce competition in each other's product spaces. This increased differentiation is reflected in higher fluidity. Importantly, in this limited context, higher fluidity does not indicate greater competition, but instead reflects greater exertion of market power through product differentiation.

Hypothesis 1C. *Under the product differentiation channel, an increase in common ownership between a firm and its product market competitors has a positive effect on fluidity.*

2.2. Common Ownership and Product Market Fluidity: The Impact of Moderating Factors

It is unlikely for any one channel to prevail across a broad set of industries with heterogeneous economic environments. These channels hinge on certain assumptions about industry structure and firms' production processes. For the efficiency channel to be a dominant economic determinant of product market dynamism, industries should possess a sufficient level of spillover opportunities between rivals (López and Vives 2019). The market power channels of common ownership, on the other hand, are expected to be more prevalent in industries in which firms can readily extract monopolistic rents.

We start by analyzing two industry attributes, concentration and technology proximity, which influence the efficiency-promoting effects of common ownership. In the López and Vives (2019) model, common owners internalize the externality from rivals' technologies with strong spillover effects that can lead to greater investments at the aggregate level. The model predicts that there are smaller potential benefits of common ownership in concentrated industries from rivals' investments because the limited number of industry participants reduces the scope of technology spillovers, thereby weakening the investment-promoting effect of common ownership. In addition, an important benefit of common ownership is economies of scale in information production (Kacperczyk et al. 2005), which are greatly curtailed in industries with only a few peer firms. Therefore, we predict that industry concentration has a negative effect on the impact of a positive shock to common ownership on product market dynamism.

Hypothesis 2A. *Under the efficiency channel, industry concentration negatively influences the impact of an increase in common ownership on fluidity.*

Gutiérrez and Philippon (2017) find that increased industry concentration reduces corporate investments and innovation because of decreased competition. As such, under the quiet life channel, industry concentration has a dampening influence on the impact of a positive shock to common ownership on fluidity because firms with increased common ownership can more easily collude to compete less aggressively in each other's product spaces.

Hypothesis 2B. *Under the quiet life channel, industry concentration has a dampening effect on the impact of an increase in common ownership on fluidity.*

In contrast, the product differentiation channel suggests that, in a more concentrated industry, firms exposed to the common ownership shock can more easily collude to lessen product market competition through greater product differentiation, thereby increasing fluidity.

Hypothesis 2C. *Under the product differentiation channel, industry concentration positively influences the impact of an increase in common ownership on fluidity.*

Further, López and Vives (2019) build on the findings of Bloom et al. (2013) to show that the knowledge spillover between firms owned by common owners is greater if rivals' technologies are more relevant to the focal firm's own technology. By internalizing the negative externality from involuntary knowledge spillovers to rivals, common ownership reduces portfolio firms' incentives to free-ride on their rivals' investments in similar technology. Thus, firms with higher common ownership are more likely to increase investments when their rivals possess more similar production technology. In sum, the relation between common ownership and product market threats is positively affected by the technology similarity between firms within an industry.

Hypothesis 3. *Under the efficiency channel, industry technology proximity has a positive effect on the impact of an exogenous shock to common ownership fluidity.*

We next analyze the moderating effects of two additional industry attributes: entry and consumer switching costs. We first consider the moderating effect of entry costs. Under the efficiency channel, an increase in common ownership increases industry spillover opportunities and results in greater product market dynamism. Higher (lower) entry costs, however, make it harder (easier) to take advantage of these industry spillover opportunities because of the larger (smaller) required investments, thereby leading to lower (higher) product market dynamism. This is because entry costs lower operating flexibility and reduce the expected profitability of starting new projects (Ovtchinnikov 2010). As a consequence, business dynamism, such as the entrance of new or existing firms in a product market, is greatly suppressed in regimes with high entry costs (Klapper et al. 2006).

Hypothesis 4A. *Under the efficiency channel, entry costs have a negative effect on the impact of an increase in common ownership on fluidity.*

We build our market power hypotheses related to entry costs on recently developed multiproduct oligopoly models that allow firms to choose both price and product offerings in equilibrium (see, e.g., Berry and Waldfogel 2001, Sweeting 2010, Fan 2013, Mazzeo et al. 2018, Wollmann 2018, Fan and Yang 2020). The market power channel predicts that concentrated ownership because of common owners can soften price competition and lead to drastic increases in markup. In theoretical models of multiproduct industries, however, such high profits in that product niche also attract rival firms to introduce new products and increase competition. Mazzeo et al. (2018) show that high fixed costs can deter potential rivals from rolling out new products in markets in which the ownership of incumbents become more concentrated, for example, through mergers. Thus, higher (lower) entry costs help (hurt) incumbents' ability to forestall entry in markets with high margins by new or existing rivals. Therefore, higher entry costs can exacerbate the decline in product market dynamism when firms with higher common ownership aim to obtain quasi-monopoly rents through a quiet life.

Hypothesis 4B. *Under the quiet life channel, entry costs have a dampening influence on the impact of an increase in common ownership on product market fluidity.*

Additionally, higher (lower) entry costs by discouraging (encouraging) entry increase (decrease) the ability to collude but also make it costlier (cheaper) for rival firms to make the investments to earn rents by differentiating their products. Since entry costs impact the ability to collude and the cost of colluding in opposing directions, the product differentiation channel has no clear-cut prediction about the moderating effect of entry costs on the impact of an exogenous increase in common ownership on product market fluidity.

Finally, we explore how common ownership interacts with switching costs between products for consumers, a key factor in competition with differentiated products (Klemperer 1987, 1995). Firms whose consumers face lower switching costs are more likely to have similar products and, therefore, employ similar production technologies. Consequently, the efficiency hypothesis suggests that firms are better positioned to take advantage of increased spillover opportunities that arise from higher common ownership if consumer switching costs are lower.

Hypothesis 5A. *Under the efficiency channel, consumer switching costs have a negative effect on the impact of an increase in common ownership on fluidity.*

Classic economic theories suggest that firms are less able to extract monopolistic rents when consumers face lower switching costs between products. Stated differently, firms have greater ability to increase prices without losing existing customers when switching costs are higher. If common owners behave as quasi-monopolists, their degree of monopoly power increases with consumer demand inelasticity. The ability to exercise market power from increased common ownership, thus, is stronger (weaker) in environments in which consumer switching costs are higher (lower).

Under the quiet life channel, firms with high common ownership have little incentive to alter their existing product portfolio in such locally monopolistic markets. Therefore, in response to a positive shock to common ownership, we expect firms to compete less (more) aggressively with each other when switching costs are higher (lower).

Hypothesis 5B. *Under the quiet life channel, consumer switching costs have a negative effect on the impact of an increase in common ownership on fluidity.*

We obtain the opposite prediction under the product differentiation channel because firms tend to differentiate their products more to earn quasi-monopoly rents when consumer switching costs are higher.

Hypothesis 5C. *Under the product differentiation channel, consumer switching costs have a positive effect on the impact of an increase in common ownership on fluidity.*

3. Data and Variables

3.1. Data Sources

We extract institutional ownership information from Thomson Reuters Institutional (13F) Holdings database (TR 13F). This ownership information is then matched to the Compustat/Center for Research in Security Prices (CRSP) merged database using the eight-digit historical Committee on Uniform Securities Identification Procedures (CUSIP) number from CRSP. Our analysis only includes U.S. domestic stocks with a CRSP share code of either 10 or 11. Fluidity and other text-based industry-level variables are downloaded from the Hoberg–Phillips Data Library (http://hobergphillips.tuck.dartmouth.edu/). The library contains fluidity (Hoberg et al. 2014), text-based network industry classifications (TNIC) (Hoberg and Phillips 2016), and the TNIC-level Herfindahl–Hirschman index (HHI). Finally, we obtain financial information from Compustat and CRSP.

3.2. Product Market Fluidity and Text-Based Industry Classifications

Our measure of the product market threats faced by a firm is the product market fluidity variable (*Fluidity*) developed by Hoberg et al. (2014). Fluidity is defined as the distance between two vectors: a vector of the firm's product description language and the other vector aggregating the changes in its rivals' product description languages. Fluidity incorporates the dynamics in the product attributes of a focal firm and its closely related rival firms. A firm faces a more fluid product market if its competitors take more actions in the firm's product space.

More formally, fluidity (*Fluidity*) is computed as the cosine similarity between vectors $N_{i,t}$ and normalized $D_{t-1,t}$ as given:

$$Fluidity_{i,t} = \left\langle N_{i,t} \cdot \frac{D_{t-1,t}}{||D_{t-1,t}||} \right\rangle, \tag{1}$$

where $N_{i,t}$ is firm i's own product description word usage vector and $D_{t-1,t}$ is the aggregate change vector of product description words of firm i's competitors. Fluidity ranges between zero and one because it is a cosine similarity measure between two normalized vectors. It is worth noting the variable fluidity capture the dynamics in the product descriptions of a firms' competitors (the element $D_{t-1,t}$). Therefore, product market fluidity can vary significantly even if a firm does not update its product description between year $t-1$ and t (i.e., $N_{i,t-1} = N_{i,t}$).

Also, fluidity should be distinguished from average product similarity between the focal firm and other firms in the same industry (*AvgSimilarity*). We illustrate this point in Online Appendix A. We start with the example in Hoberg et al. (2014) and use it as the benchmark to compare with other product market strategies. We first extend their example to illustrate that a higher value for fluidity can be associated with either a decrease or increase in average product similarity. We then construct two additional extreme examples. In the first example, firms do not change their product descriptions, thereby exemplifying a quiet life. Clearly, fluidity is zero and average product similarity remains unchanged. In the second example, firms coordinate changes to their product descriptions such that there is no overlap with the product spaces of other firms in the industry. In this extreme form of product differentiation, fluidity goes up, but average product similarity goes down to zero.

Hoberg et al. (2014) show that firms with a higher fluidity are less likely to pay dividends and have higher cash balances, suggesting that these firms pursue financial flexibility to respond more dynamically to changing product markets. In Online Appendix A, we also estimate an ordinary least squares (OLS) regression model to study the relation between fluidity and several firm characteristics. We document that fluidity is higher for firms that are younger with higher growth opportunities and more volatile stock returns. Additionally, firms in more concentrated industries tend to have a lower fluidity. These results support the interpretation of fluidity made in Hoberg et al. (2014) as a proxy for product market dynamism and threats.

Hoberg et al. (2014) compute fluidity using product market competitors classified by TNIC (Hoberg and Phillips 2010, 2016). TNIC is built on a textual analysis of each firm's 10-K product description languages to quantify the product similarity between firms to determine a firm's product market competitors. Hoberg and Phillips (2016) show that TNIC more accurately captures changes in product market rivalry over time at the firm level. It also fits well with regression models that include both year and firm fixed effects. We use TNIC to construct all industry-specific variables. For some tests that can benefit from traditional types of industry classifications, we use the fixed industry classifications (FIC) of Hoberg and Phillips (2016) built on the same textual analysis of 10-Ks used in TNIC but which maintain a fixed group of industries and transitivity of competitors. Finally, following Hoberg and Phillips (2010), we use the size (in words) of a firm's 10-K product description to study whether common ownership is associated with new product development.

3.3. Measures of Common Ownership

We measure common ownership at the fund family level rather than at the fund level. There are a few reasons for this approach. First, the board of directors tend to be the same across funds in a fund family (https://www.ici.org/doc-server/pdf%3Abro_mf_directors.pdf). Second, mutual funds typically make their voting rights available to their fund family's central proxy voting office (Rothberg and Lilien 2006). Third, these offices often actively engage with portfolio firms in an attempt to increase their values. Maximizing the value of portfolio firms maximizes assets under management and, thus, aligns the incentives of the fund family with the incentives of the investors (Gaspar et al. 2006). See, for example, Azar et al. (2018, 2022a) for similar arguments.

We use three measures of common institutional ownership among industry competitors in our panel regression models. The first measure is *MHHID*, originally developed by O'Brien and Salop (2000) and used in Azar et al. (2018). Our second measure, *Cross*, follows the measure used in Hansen and Lott (1996). Finally, our third measure is the *GGL* measure developed by Gilje et al. (2020). The details regarding how we construct these three measures are provided in Appendix A. Our DID analyses that exploit mergers between financial institutions as exogenous shocks to common ownership are, however, independent of the specific choice of common ownership measure. Nevertheless, we do use these measures to validate our DID experimental design.

3.4. Descriptive Statistics

Our sample selection process begins with firm–year observations from the Compustat/CRSP merged database with nonmissing fluidity. Further, we require that sample firms have nonmissing and positive total assets and sales and nonmissing key control variables used in our tests: Tobin's Q, return on assets (ROA), book leverage, cash, stock return volatility, and HHI. After matching this sample to the 13F ownership information, we obtain 11,335 unique firms and 85,400 firm–year observations between 1997 and 2017. The 13F filings report institutional ownership for 98.6% of our final sample. For the firm–year observations with missing ownership information, we assign zero to all common ownership-related variables. Excluding these observations produces almost identical results.

Table 1 shows descriptive statistics for the variables used in our tests. Fluidity and average product similarity are presented in percentage points to improve the readability of the results. The mean (median) value of fluidity is 7.26% (6.61%) and the mean (median) value of average product similarity is 3.73% (2.81%). The length of product description has a mean of 4,542 words and a median of 3,856 words. Investments as a proportion of total assets have a mean of 14% and a median of 7.9%. Next, the mean (median) value of *MHHID* and *HHI* are 0.127 (0.051) and 0.231 (0.14), respectively. Further, the mean (median) of *Cross* and *GGL* are 1.083 (0.161) and 0.841 (0.439). Other firm characteristics are comparable to those in Hoberg et al. (2014).

Table 1. Descriptive Statistics

	Observations	Mean	Standard deviation	P25	P50	P75
Dependent variable						
Fluidity, %	85,400	7.263	3.578	4.592	6.613	9.328
AvgSimilarity, %	85,400	3.734	3.604	1.890	2.810	4.304
Product Description (in 1,000 s)	85,400	4.542	2.846	2.488	3.856	5.879
Investments	85,400	0.140	0.184	0.022	0.079	0.181
Common ownership measure						
MHHID	85,400	0.127	0.163	0.000	0.051	0.201
Cross	85,400	1.083	2.253	0.007	0.161	0.991
GGL	85,400	0.841	1.336	0.052	0.439	1.067
TNIC-level independent variables						
HHI	85,400	0.231	0.227	0.074	0.140	0.305
TechProximity	19,697	0.242	0.224	0.040	0.196	0.389
PPE (in \$ billions)	85,205	0.514	1.127	0.022	0.082	0.406
TotalProdSimilarity	83,832	7.426	15.279	0.272	1.339	5.620
Firm-level control variables						
Assets (in \$ billions)	85,400	6.647	55.528	0.112	0.505	2.136
Age	85,400	18.000	14.537	7	13	24
Tobin's Q	85,400	1.995	1.716	1.045	1.381	2.188
ROA	85,400	0.048	0.204	0.019	0.085	0.149
Leverage	85,400	0.226	0.222	0.026	0.172	0.360
Cash	85,400	0.189	0.224	0.026	0.089	0.274
Return volatility, %	85,400	3.581	2.176	1.977	3.007	4.565

Notes. This table reports the descriptive statistics for the variables used in this paper. Our sample consists of 85,400 firm–year observations between 1997 and 2017. See Appendix A for the complete list of variable definitions. All continuous and unbounded variables are winsorized at the 1st and 99th percentiles or log-transformed.

4. The Effects of Common Ownership on Product Market Threats

4.1. Panel Regression Results

This section examines the net effect of common institutional ownership on product market fluidity in a panel regression setting. Specifically, we estimate the following regression model:

$$Fluidity_{it} = \beta \times Common\ Ownership_{it-1} + Controls_{it-1} + Year\ FE + Firm\ FE + \epsilon_{it}, \qquad (2)$$

where $Fluidity_{it}$ (in percentage points) is the measure of product market threats faced by firm i in year t. The variable, $Common\ Ownership_{it-1}$ is one of the three measure of common ownership (*MHHID*, *Cross*, and *GGL*) between firm i and its product market competitors in year $t-1$. We control for several firm characteristics that can be correlated with fluidity as discussed in Hoberg et al. (2014): firm size and age, Tobin's Q, ROA, leverage, cash, and annual stock return volatility. These variables are defined in Appendix A. All regression models include year and firm fixed effects to control for any unobservable time trends and time-invariant heterogeneity in firm characteristics. In some models, we include multiplicative industry × year fixed effects to absorb economic or regulatory shocks that may affect a specific group of firms simultaneously. When constructing industry × year fixed effects, we use time-invariant text-based FIC of Hoberg and Phillips (2016) to be consistent with the TNIC definition.

Table 2 reports our OLS estimates. The odd columns report models with firm and year fixed effects; the even columns report firm and industry × year fixed effects. Following Azar et al. (2018), standard errors are two-way clustered by year and firm or by year and industry if industry × year fixed effects are included in the regression model. The two-way clustered standard errors correct for both the serial correlation in error terms within a firm and the cross-sectional correlation in error terms across firms in the same year.

In the first two models of Table 2, the coefficient on *MHHID* is positive and statistically significant at the 1% level with and without industry × year fixed effects. Note that we compute *HHI* at the TNIC-level, whereas the multiplicative industry × year fixed effect is defined at the FIC-level industry. Hence, the coefficient on *HHI* is still estimated.

The effect is also economically meaningful. For example, column (2) indicates that a one standard deviation increase in *MHHID* (=0.163) is associated with a 0.272 (=0.163 × 1.668) percentage point increase in fluidity (about 7.6% of fluidity standard deviation and a percentage change of 3.7% relative to its mean value of 7.263%). In columns (3) and (4), we show that the positive relationship between common ownership and fluidity remains positive and statistically significant at the 1% level when we consider the *Cross* measure. Finally, we

Table 2. Product Market Fluidity and Common Institutional Ownership

Common ownership measure	*MHHID*		*Cross*		*GGL*	
Dependent variable: *Fluidity*	(1)	(2)	(3)	(4)	(5)	(6)
Common ownership	1.079***	1.668***	0.142***	0.098***	0.055	0.078*
	(3.72)	(7.82)	(4.78)	(4.19)	(1.28)	(1.94)
HHI	−1.201***	−0.926***	−1.154***	−0.975***	−1.259***	−1.022***
	(−14.04)	(−7.59)	(−12.81)	(−7.41)	(−14.67)	(−7.34)
Ln(Assets)	0.396***	0.256***	0.355***	0.243***	0.405***	0.274***
	(9.68)	(6.67)	(9.08)	(6.51)	(9.95)	(6.63)
Ln(Age)	−0.711***	−0.488***	−0.778***	−0.518***	−0.734***	−0.508***
	(−6.75)	(−5.69)	(−7.67)	(−5.76)	(−7.00)	(−5.96)
Tobin's Q	0.075***	0.049***	0.069***	0.045***	0.074***	0.049***
	(6.16)	(4.21)	(5.63)	(3.90)	(6.25)	(4.58)
ROA	−0.831***	−0.472***	−0.836***	−0.486***	−0.849***	−0.491***
	(−6.71)	(−4.47)	(−6.95)	(−4.61)	(−6.77)	(−4.50)
Leverage	−0.209	0.009	−0.152	0.032	−0.206	0.012
	(−1.59)	(0.11)	(−1.20)	(0.36)	(−1.57)	(0.14)
Cash	0.404***	0.197*	0.403***	0.198*	0.399***	0.188*
	(4.48)	(1.88)	(4.54)	(1.91)	(4.47)	(1.75)
Return volatility	0.118***	0.043**	0.117***	0.044**	0.117***	0.044**
	(7.24)	(2.54)	(7.30)	(2.60)	(7.41)	(2.63)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	No	Yes	No	Yes	No
Industry (FIC) × Year fixed effects	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.781	0.821	0.783	0.820	0.781	0.820
Observations	85,400	85,400	85,400	85,400	85,400	85,400

Notes. This table reports OLS estimates on the relation between a firm's common institutional ownership with its industry competitors and the product market threats faced by a firm, measured by product market fluidity. The dependent variable is fluidity measured in percentage points. The main independent variable is a measure of common ownership: *MHHID* in columns (1) and (2), *Cross* in columns (3) and (4), and *GGL* in columns (5) and (6). Our measures of common ownership are computed using TNIC of Hoberg and Phillips (2016). See Appendix A for the complete list of variable definitions. Standard errors are two-way clustered by year and firm or year and industry if industry-year fixed effects are included. In our regression models, industry-year fixed effects are at the 100 FIC of Hoberg and Phillips (2016). *t*-statistics are in parentheses.
***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

find a positive association between *GGL* and fluidity. While the relation is not significant in the model with firm and year fixed effects (*t*-statistic = 1.28), it becomes statistically significant at the 10% level and economically larger when we add the interactive industry × year fixed effect to the regression model. Similar to the estimates of the *MHHID* models, both *Cross* and *GGL* have an economically significant relation with fluidity. Overall, our OLS estimates show that common ownership is positively related to the firm's product market threats. These findings provide initial support for the efficiency channel. They, however, also lend support to the product differentiation channel.

4.2. DID Estimation Using Institutional Mergers

Though our baseline panel results show a robust positive relation between common ownership and product market fluidity, they do not allow us to make any causal claims. One potential endogeneity concern is that the OLS regression results are simply attributable to selection; that is, institutional investors have a tendency to jointly hold firms with certain product market attributes rather than actually causing changes in these firms' product market strategies. Another potential endogeneity concern is that some unobservable factor such as similar technological advancements by competing firms may simultaneously affect institutional cross-holdings of competitors and subsequent product market threats from these competitors.

To establish causality from common ownership to fluidity, we exploit mergers between financial institutions as exogenous shocks that can increase common ownership between product market competitors. Following He and Huang (2017) and Lewellen and Lowry (2021), we estimate DID regression models using multiple merger events. Our empirical strategy allows us to alleviate the selection issue because we can more definitively relate changes in product market strategies to exogenous changes in common ownership. In addition, mergers between financial institutions are likely to be driven by economic factors that are largely unrelated to their portfolio holdings and, in turn, the product market strategies of their portfolio firms (Huang 2014). Consequently, our experimental setting also alleviates the omitted variable issue because these mergers are likely to create exogenous variation in common ownership that is independent of the product market strategies of their portfolio firms.

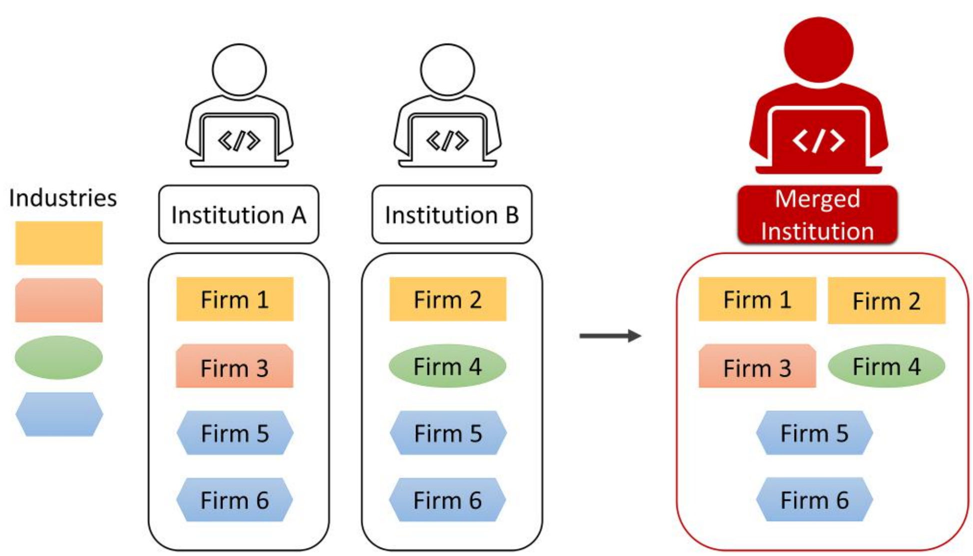
Lewellen and Lowry (2021) argue that these merger events provide the most credible identification strategy in common ownership research although some merger events during the financial crisis could be potentially contaminated by other factors. To address this concern, we collect and extensively analyze all institutional mergers from Security Data Company (SDC) Platinum and conduct our empirical analysis based on two samples: (1) the union of merger events (All) reported in He and Huang (2017) and Lewellen and Lowry (2021) and (2) the preceding set of mergers that occur outside of the 2008–2009 period (Recommended) as recommended by Lewellen and Lowry (2021). During our sample period, we identify 27 and 21 institutional mergers for the All and Recommended events, respectively. The Recommended mergers contain 18 events from Lewellen and Lowry (2021) and three events from He and Huang (2017) that are not included in Lewellen and Lowry (2021). We analyze the information about the latter mergers from SDC and Google and conclude that they are valid merger events.

For each merger event, we define treatment and control groups using a similar approach described in He and Huang (2017) and Lewellen and Lowry (2021). First, we identify firms held by one of the merging institutions (acquirer or target) with a 0.5% or greater ownership one quarter before the announcement of the merger, whereas the other merging institution does not have a 0.5% or greater ownership at the same time. Among these firms, a firm is assigned to the treatment group if the merging institution with less than 0.5% ownership in the firm has a 0.5% or greater ownership of any of the firm's competitors. On the other hand, a firm is assigned to the control group if the merging institution with less than 0.5% ownership in the firm does not have a 0.5% or greater ownership of any of the firm's competitors. This strategy controls for institution-specific factors such as managerial styles or abilities. Finally, both merging institutions may already have a 0.5% or greater ownership in a firm and its competitors before the merger. We note that the merger still increases the extent of common ownership in these cases and, therefore, include them in the treatment group.[3] Our DID sample consists of the treatment and control groups of firm–year observations during (−2, +2) or (−3, +3) years relative to each merger announcement year.[4]

Figure 1 illustrates an example of changes in common institutional ownership after a merger between two institutions A and B. Institution A holds at least 0.5% ownership of firms 1, 3, 5, and 6 before the merger. Institution B holds a 0.5% ownership of firms 2, 4, 5, and 6 before the merger. Firms 1 and 2 are included in the treatment group as they are product market competitors (i.e., operate in the same industry) before the merger and become cross-held by the merged institution after the merger. Firms 3 and 4 are included in the control group as they have no product market competitors cross-held after the merger. Firms 5 and 6 are also included in the treatment group as these firms experience a substantial increase in the extent of common ownership even though they are held by both institutions before the merger.

To identify the strength of influence common ownership imposes on firms' competition strategy, we consider

Figure 1. (Color online) Illustration of Changes in Common Institutional Ownership After a Merger Between Financial Institutions



Notes. This figure illustrates an example of changes in common institutional ownership after a merger between two institutions A and B. Institution A holds at least 0.5% ownership of firms 1, 3, 5, and 6 before the merger. Institution B holds at least 0.5% ownership of firms 2, 4, 5, and 6 before the merger. Firms 1 and 2 are included in the treatment group as they are product market competitors (i.e., operate in the same industry) before the merger and become cross-held by the merged institution after the merger. Firms 3 and 4 are included in the control group as they have no product market competitors cross-held after the merger. Firms 5 and 6 are also included in the treatment group as these firms experience a substantial increase in the level of common ownership even though they are held by both institutions before the merger.

two definitions of the treatment group based on the length of institutional holding periods. Given that our hypotheses center on the product market effects of common ownership, it is reasonable to expect that long-term institutional owners have a more identifiable effect on product market strategies and product offerings compared with investors who churn their portfolios frequently. Therefore, we construct our empirical tests around two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted), imposing an additional condition to (i) that such common ownership is anticipated to last four or more out of 12 quarters after the mergers.[5] Our prediction model for the persistence of common ownership after institutional mergers is based on diversification needs and regulatory concerns of institutional investors. To ensure that our model does not capture ex post realized outcomes from the mergers, we estimate our model using only a set of merger events that occurred in the 10-year period before the announcement of each merger.[6] For the regression models using (ii), we drop any treated firms included in (i) but not in (ii).[7] We believe that the results based on the 4+ Quarters (Predicted) definition capture the influence of likely long-term common owners.

Our DID regression model is as follows:

$$Y_{it} = \beta \times Post \times Treatment + Merger \times Firm\ FE + Merger \times Year\ FE + Controls_{it-1} + \epsilon_{it}, \tag{3}$$

where *Treatment* is an indicator variable that equals one if the firm's common ownership structure with competitors is affected by the merger and zero otherwise. Following the previous discussion on the institutional holding period, we report results using all common owners (Any Quarters) as well as those common owners that are likely to hold the stock for at least 4 out 12 quarters after the mergers (4+ Quarters (Predicted)). *Post* is an indicator variable that equals one if the observation is made after the merger and zero otherwise. Following Gormley and Matsa (2011), treatment and control groups identified for each merger event are stacked in our regression sample. As a result, a firm-year observation can be in either the treatment or control group for different merger events. With merger × year and merger × firm fixed effects, our coefficients are solely identified by the within-firm merger variation in variables. We use two different event windows, (−2, +2) and (−3, +3) years, and two sets of merger events, All and Recommended.[8] The sample size from Recommended mergers is about 60% of All mergers because a few major mergers that occurred during the financial crisis (e.g., BlackRock's acquisition of BGI) are dropped from the analysis. The control variables are the same as those used in the OLS regressions reported in Table 2.[9]

In Table 3, we check the validity of our DID research design by examining whether institutional mergers are indeed associated with an increase in the common ownership between competitors. Panels A–C report results on the postmerger changes in the three common ownership measures: *MHHID*, *Cross*, and *GGL*, respectively. We document that all measures of common ownership increase after institutional mergers at the statistical significance levels of 1% except for the estimates on *MHHID* using the All mergers sample. We, therefore, conclude that the institutional mergers offer a valid quasi-natural experiment that creates an exogenous and positive shock to common institutional ownership between competitors. These results form the basis for using the DID estimation methodology throughout our study. This methodology not only provides more reliable coefficients that suffer less from endogeneity issues, but the coefficients are also largely independent of the choice of common ownership measures.

4.3. Baseline DID Results

We now study the relation between common ownership and fluidity using institutional mergers as our experimental setting. Panel A of Table 4 reports DID estimation results based on Equation (5) with fluidity as the main dependent variable. Following the previous section, we report models studying two sets of merger events (All and Recommended) under two alternative definitions of the treatment group based on holding periods (Any Quarters and 4+ Quarters (Predicted)). In all regression models, the coefficient on *Post* × *Treatment* is always positive and statistically significant at least at the 5% level except for the estimates in the Recommended merger sample in the (−2, +2) window (*t*-statistic = 1.47). These results support the hypothesis that common ownership promotes competition in firms' product spaces. The magnitude of the statistically significant coefficients shows that, on average, the treatment group experiences a magnitude of 0.082%–0.156% increase in fluidity (a percentage change of 1.3%–2.1% relative to the mean level of fluidity).

With the inclusion of merger × year and merger × firm fixed effects, the results show that our DID regression models capture within-firm changes in product market behavior rather than spurious correlation induced by cross-firm variations because of possible portfolio churning by institutional investors. Overall, our baseline DID estimation results provide evidence that, on average, common ownership has a causal and net positive effect on the product market threats faced by the firm. As per our hypotheses, these baseline fluidity results are consistent with both the efficiency and product differentiation channels but inconsistent with the quiet life channel.

Table 3. How Does Common Institutional Ownership Change After Institutional Mergers?

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
				Panel A. Dependent variable: *MHHID*				
Post × *Treatment*	0.002	0.003	0.014***	0.016***	−0.000	0.002	0.012***	0.015***
	(1.14)	(1.50)	(5.96)	(6.57)	(−0.03)	(0.67)	(4.51)	(5.28)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.761	0.749	0.753	0.738	0.766	0.753	0.758	0.742
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
				Panel B. Dependent variable: *Cross*				
Post × *Treatment*	0.437***	0.559***	0.574***	0.685***	0.541***	0.662***	0.691***	0.792***
	(15.89)	(17.67)	(17.62)	(19.09)	(16.90)	(18.47)	(18.77)	(19.90)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.848	0.821	0.841	0.822	0.853	0.824	0.843	0.824
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
				Panel C. Dependent variable: *GGL*				
Post × *Treatment*	0.079***	0.098***	0.076***	0.085***	0.077***	0.090***	0.069***	0.074***
	(7.93)	(9.02)	(6.62)	(6.66)	(6.97)	(7.44)	(5.54)	(5.29)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.741	0.709	0.732	0.701	0.749	0.719	0.743	0.711
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474

Notes. This table validates our DID model by testing the impact of institutional mergers on three measures of common ownership: *MHHID*, *Cross*, and *GGL*. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted), imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models using (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. The main independent variable is the interaction term between *Post* and *Treatment* that estimates the impact of institutional mergers on fluidity. See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.
***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

To differentiate whether the positive "average" effect of common ownership is due to the efficiency or product differentiation channel, we estimate the same DID regression model specified in Equation (6) with *AvgSimilarity* as the dependent variable. The results from these regressions are reported in panel B of Table 4. We find that coefficients on *Post* × *Treatment* are significantly positive at least at 5% in all the regressions. This finding is inconsistent with the product differentiation channel. Thus, our baseline results for *Fluidity* and *AvgSimilarity* together suggest that exogenous shocks to common ownership increase product market threats, which is consistent with the efficiency channel as the dominant motive for the average firm in our sample.

We hypothesized earlier that the impact of common ownership on product market threats is likely affected by industry attributes. As a preliminary examination of possible heterogeneity in this relation, we use the same DID regression setting to examine the impact of common ownership on product market fluidity for each Fama–French (FF) 49 industry. Specifically, we additionally include 48 (excluding the base industry) industry dummies (*FF Dummy*) and their interaction terms with DID indicators in the regression models: *FF Dummy*, *Post* × *FF Dummy*, *Treatment* × *FF Dummy*, and *Post* × *Treatment* × *FF Dummy*. We estimate this regression 49 times, changing the base industry each time to obtain the coefficient on *Post* × *Treatment* for each (base) industry.

In panel C of Table 4, we report the estimates for industries in which the interaction term between *Post* and *Treatment* is statistically significant at least at the 10% level from the DID models with 4+ Quarters (Predicted) and Recommended mergers estimated over either the (−2, +2) or (−3, +3) window. We find that common ownership significantly increases product market threats in the drug, rubber, coal, telecom, hardware, boxes, and bank industries. On the other hand, common ownership significantly decreases product market threats in the

Table 4. DID Estimation Using Institutional Mergers

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
Panel A. Product market fluidity: Baseline results								
Post × Treatment	0.121***	0.127***	0.062	0.138***	0.145***	0.139***	0.095**	0.168***
	(3.48)	(3.49)	(1.47)	(3.34)	(3.65)	(3.36)	(1.99)	(3.61)
HHI	−0.568***	−0.791***	−0.554***	−0.715***	−0.575***	−0.769***	−0.524***	−0.705***
	(−7.33)	(−10.40)	(−6.13)	(−8.34)	(−6.91)	(−9.35)	(−5.38)	(−7.73)
Ln(Assets)	0.395***	0.390***	0.354***	0.369***	0.370***	0.357***	0.353***	0.353***
	(10.47)	(11.01)	(8.13)	(9.65)	(8.91)	(9.20)	(7.62)	(8.76)
Ln(Age)	−0.762***	−0.816***	−0.917***	−0.861***	−0.815***	−0.897***	−0.919***	−0.866***
	(−7.29)	(−8.58)	(−7.68)	(−8.33)	(−7.24)	(−8.81)	(−7.07)	(−7.92)
Tobin's Q	0.081***	0.071***	0.068***	0.065***	0.084***	0.072***	0.071***	0.064***
	(5.57)	(5.11)	(4.00)	(4.20)	(5.10)	(4.58)	(3.77)	(3.80)
ROA	−0.690***	−0.848***	−0.525**	−0.619***	−0.578***	−0.833***	−0.427*	−0.558***
	(−3.99)	(−5.28)	(−2.54)	(−3.45)	(−2.89)	(−4.45)	(−1.83)	(−2.82)
Leverage	−0.310**	−0.266**	−0.189	−0.197	−0.235*	−0.215	−0.146	−0.163
	(−2.42)	(−2.16)	(−1.25)	(−1.51)	(−1.67)	(−1.59)	(−0.89)	(−1.17)
Cash	0.201	0.278**	−0.027	0.034	0.196	0.297**	0.034	0.083
	(1.50)	(2.14)	(−0.16)	(0.24)	(1.31)	(2.08)	(0.19)	(0.54)
Return volatility	0.179***	0.188***	0.112***	0.152***	0.183***	0.187***	0.126***	0.160***
	(12.09)	(13.64)	(6.85)	(11.10)	(11.70)	(12.74)	(7.11)	(10.89)
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.821	0.816	0.817	0.817	0.818	0.815	0.812	0.813
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel B. Average product similarity: Baseline results								
Post × Treatment	0.208***	0.234***	0.137***	0.136**	0.153***	0.197***	0.127**	0.136**
	(3.23)	(3.24)	(2.72)	(2.47)	(2.63)	(3.06)	(2.33)	(2.27)
HHI	−0.141	−0.081	0.050	0.097	−0.099	0.037	0.070	0.201
	(−0.94)	(−0.56)	(0.28)	(0.48)	(−0.56)	(0.22)	(0.34)	(0.90)
Ln(Assets)	0.041	0.041	−0.012	−0.009	0.031	0.027	−0.019	−0.030
	(0.93)	(0.87)	(−0.31)	(−0.20)	(0.60)	(0.48)	(−0.43)	(−0.57)
Ln(Age)	−0.093	−0.154	0.018	−0.039	−0.027	−0.141	0.098	0.004
	(−0.57)	(−0.86)	(0.11)	(−0.24)	(−0.14)	(−0.66)	(0.57)	(0.02)
Tobin's Q	0.022*	0.026**	0.012	0.021	0.021	0.026	0.015	0.024
	(1.87)	(1.96)	(0.91)	(1.30)	(1.42)	(1.56)	(0.95)	(1.22)
ROA	−0.112	−0.126	0.064	−0.078	−0.198	−0.183	0.025	−0.125
	(−0.75)	(−0.79)	(0.44)	(−0.46)	(−1.06)	(−0.93)	(0.15)	(−0.63)
Leverage	−0.141	−0.115	0.019	0.010	−0.107	−0.108	0.056	0.022
	(−1.18)	(−0.91)	(0.14)	(0.07)	(−0.77)	(−0.74)	(0.37)	(0.14)
Cash	−0.026	−0.014	0.092	0.041	−0.020	−0.003	0.105	0.043
	(−0.22)	(−0.12)	(0.72)	(0.29)	(−0.14)	(−0.02)	(0.71)	(0.26)
Return volatility	0.026*	0.025*	−0.015	−0.007	0.017	0.020	−0.006	0.001
	(1.87)	(1.77)	(−0.79)	(−0.44)	(1.14)	(1.26)	(−0.32)	(0.03)
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.744	0.714	0.716	0.672	0.734	0.703	0.715	0.669
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474

	Event window		Median value of industry attributes			
Fama-French industry	(−2, +2) (1)	(−3, +3) (2)	*HHI* (3)	*TechProximity* (4)	*PPE* ($ billions) (5)	*TotalProdSimilarity* (6)
Panel C. Product market fluidity: Fama–French 49 industry-level results						
2. Food	−0.281**	−0.283**	0.447	0.401	1.026	0.089
	(−2.21)	(−2.39)				
6. Toys	−0.636*	−0.346	0.415	0.400	0.060	0.138
	(−1.90)	(−1.20)				

Table 4. (Continued)

	Event window		Median value of industry attributes			
Fama-French industry	(−2, +2) (1)	(−3, +3) (2)	*HHI* (3)	*TechProximity* (4)	*PPE* ($ billions) (5)	*TotalProdSimilarity* (6)
8. Books	−0.398*	−0.429*	0.187	0.053	0.432	0.430
	(−1.84)	(−1.95)				
9. Household	−0.425***	−0.332**	0.344	0.285	0.306	0.135
	(−2.97)	(−1.98)				
13. Drug	0.362**	0.377**	0.122	0.465	0.014	14.551
	(2.31)	(2.43)				
14. Chemical	−0.393***	−0.273*	0.241	0.298	1.538	0.207
	(−2.74)	(−1.91)				
15. Rubber	0.139	0.323*	0.421	0.050	0.604	0.049
	(0.80)	(1.71)				
17. Building material	−0.540*	−0.516**	0.311	0.106	0.655	0.132
	(−1.95)	(−2.14)				
19. Steel	−0.584**	−0.519*	0.197	0.064	0.713	0.298
	(−2.08)	(−1.91)				
20. Fabricated product	−0.430*	−0.167	0.358	0.044	0.437	0.095
	(−1.68)	(−0.83)				
21. Machinery	−0.337**	−0.244	0.277	0.186	0.408	0.173
	(−2.44)	(−1.59)				
23. Auto	−0.329**	−0.333*	0.319	0.216	0.821	0.178
	(−2.01)	(−1.84)				
29. Coal	1.334*	2.267**	0.056	0.095	2.145	4.771
	(1.70)	(2.45)				
30. Oil	−0.419**	−0.427**	0.082	0.276	0.976	3.744
	(−2.23)	(−2.27)				
32. Telecom	0.293	0.430**	0.120	0.328	0.522	2.641
	(1.34)	(1.97)				
33. Personal service	−0.396	−0.606***	0.185	0.000	0.184	0.667
	(−1.34)	(−2.70)				
34. Business service	−0.249*	−0.097	0.261	0.229	0.074	0.396
	(−1.67)	(−0.65)				
35. Hardware	0.468**	0.267	0.141	0.210	0.056	1.837
	(2.14)	(1.33)				
37. Chips	0.198*	0.248**	0.128	0.194	0.080	2.336
	(1.73)	(2.41)				
39. Paper	−0.304*	−0.296	0.307	0.472	1.493	0.149
	(−1.76)	(−1.47)				
40. Boxes	0.220	0.553**	0.196	0.318	2.424	0.194
	(0.98)	(2.04)				
45. Bank	0.788**	1.031***	0.068	0.000	0.042	50.236
	(2.46)	(3.28)				

Notes. This table presents the relation between common institutional ownership and product market strategies using DID models that exploit a set of mergers between financial institutions. Panel A reports the results from the DID models with fluidity as the dependent variable using the full sample. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted), imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. The main independent variable is the interaction term between *Post* and *Treatment* that estimates the impact of institutional mergers on fluidity. Panel B reports the estimates from the same DID models with average product similarity as the dependent variable. Panel C reports the DID estimates on fluidity and median value of industry attributes for a set of Fama–French 49 industries in which the interaction term between *Post* and *Treatment* is statistically significant at least at the 10% level in either window on tests with Any Quarters and Recommended mergers. Columns (1) and (2) report the estimated coefficients. Columns (3)–(6) report the median value of industry concentration (*HHI*), technology proximity (*TechProximity*), entry costs (*PPE*) ($ billions), and consumer switching costs (*TotalProdSimilarity*). See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.

***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

food, toys, household, chemical, building material, steel, electric equipment, auto, gold, oil, personal service, software, and paper industries.

This panel also reports median values for industry characteristics such as industry concentration (proxied by *HHI*), industry technological similarity (proxied by

TechProximity), industry entry costs (proxied by *PPE*), and consumer switching costs (proxied by *TotalProdSimilarity*) to provide some intuition for these findings. We discuss the construction of these variables in Section 4.4 and Appendix A. For example, based on the descriptive statistics for our DID sample (Online Appendix Table 1), the drug industry is characterized by relatively low industry concentration, high technological proximity, low entry costs, and low consumer switching costs. In contrast, the steel industry is characterized by relatively high concentration, low technological proximity, high entry costs, and high consumer switching costs. Given our earlier theoretical arguments, it is not surprising that the spillover effects outweigh (are outweighed by) the quiet life effects in the drug (steel) industry.[10]

4.4. Economic Mechanism

Our industry-level regression results preclude a one-size-fits-all conclusion for the impact of common ownership on product market threats. We, therefore, delve into the mechanisms underlying our results to provide evidence on the heterogeneous effects of positive shocks to common ownership on product market threats along four dimensions of industry characteristics: industry concentration, technology similarity, entry costs, and consumer switching costs.

We first examine the industry concentration dimension. Specifically, we estimate a triple difference regression model by including an additional interaction term of TNIC-level *HHI* in our DID framework. Panel A of Table 5 reports the estimated coefficients. For brevity, we only report coefficients on the double interaction term *Post* × *Treatment* and triple interaction term *Post* × *Treatment* × *HHI* (the detailed results are reported in Online Appendix Table 4). We find that the coefficients on *Post* × *Treatment* × *HHI* are negative and significant at the 1% level in all regression models. The negative coefficients on *Post* × *Treatment* × *HHI* are consistent with the efficiency channel (Hypothesis 2A) and the quiet life channel (Hypothesis 2B) but inconsistent with the product differentiation channel (Hypothesis 2C). To provide some additional intuition, as an example, the coefficient on *Post* × *Treatment* is 0.263 and the coefficient on *Post* × *Treatment* × *HHI* is −0.774 in column (3). This implies that the coefficient on *Post* × *Treatment* is positive (negative) in industries with *HHI* less (greater) than 0.340. Given that the inflection point of 0.340 is greater than the 75th percentile value of *HHI* (=0.305), the quiet life incentives only tend to dominate in highly concentrated industries. Alternatively, the investment promoting effects of common ownership are stronger in less concentrated industries (López and Vives 2019).

Next, we examine the impact of the industry-level similarity in technology portfolios on the relation between common ownership and fluidity. We estimate industry-level technology proximity by calculating the median cosine similarity between patent portfolios (Jaffe 1986) of a firm and its TNIC rivals (*TechProximity*). To construct firm-level patent portfolios, we use the patent data set from Arora et al. (2021) that identifies firm-level matches for U.S. granted patents for 1980–2015. Following Bena and Li (2014) and Li et al. (2019), we measure a firm's patent portfolio in year t as a set of patents filed between year $t-2$ and t.

Panel B of Table 5 reports the impact of technology proximity on our baseline results. For brevity, we only report the coefficients of interest (detailed results are reported in Online Appendix Table 4). We find that the coefficient of the triple-interaction term between *Post*, *Treatment*, and *TechProximity* is positive in all regression models. The coefficient is statistically significant at least at the 5% level in six out of eight models. This finding is consistent with the efficiency channel (Hypothesis 3). To again provide some additional insights, in column (3), the coefficient on *Post* × *Treatment* is −0.346 and the coefficient on *Post* × *Treatment* × *TechProximity* is 1.028. This implies that the coefficient on *Post* × *Treatment* is positive (negative) in industries with *TechProximity* greater (less) than 0.337 (which is less than the 75th percentile value of *TechProximity*). Thus, the efficiency (quiet life) channel tends to dominate in industries in which firms operate in more (less) similar technological spaces.

We then analyze how entry costs influence the product market effects of common ownership. Siegfried and Evans (1994) find that the absolute amount of capital required to build a minimum efficient scale plant (entry costs) is a deterrent to entry. Karuna (2007, pp. 282–283) defines entry costs as the "minimum level of investment (exogenous sunk cost) that must be incurred by each entrant firm to an industry prior to commencing production (i.e., set-up costs)." He proposes the use of gross (acquisition value) property, plant, and equipment for industry firms to capture the minimal level of investment needed to enter the market. We, therefore, use the natural logarithm of the median value of gross property, plant, and equipment (*PPE*) for all industry firms as a proxy for entry costs by either new entrants or existing firms to compete in each other's product spaces.

Our DID estimation results on entry costs are reported in panel C of Table 5. For brevity, we only report the coefficients of interest (detailed results are reported in Online Appendix Table 4). The estimated coefficient on *Post* × *Treatment* × Ln(*PPE*) is negative and significant at the 1% level in all regression models. These results support the efficiency channel (Hypothesis 4A) and the quiet life channel (Hypothesis 4B). Again, as an illustrative example, we find that the coefficient on *Post* × *Treatment* is 0.592 and the coefficient on *Post* × *Treatment* × Ln(*PPE*) is −0.100 in column (3). This implies that the coefficient on *Post* × *Treatment* is positive (negative) in industries with Ln(*PPE*) less (greater) than 5.92 (*PPE* of $372 million, which is less than the 75th percentile value of *PPE*).

Table 5. The Impact of Industry Concentration, Technology Proximity, Entry Costs, and Consumer Switching Costs on the Relation Between Common Ownership and Fluidity

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
Panel A. Industry concentration: *HHI*								
Post × Treatment	0.286***	0.336***	0.263***	0.368***	0.303***	0.321***	0.298***	0.406***
	(5.46)	(6.43)	(4.32)	(6.39)	(5.16)	(5.53)	(4.34)	(6.30)
Post × Treatment × HHI	−0.832***	−0.841***	−0.774***	−0.771***	−0.830***	−0.687***	−0.831***	−0.846***
	(−6.19)	(−6.44)	(−4.62)	(−4.97)	(−4.29)	(−3.80)	(−3.63)	(−4.01)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.821	0.816	0.817	0.818	0.819	0.815	0.812	0.814
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel B. Industry technology proximity: *TechProximity*								
Post × Treatment	−0.308***	−0.317***	−0.346***	−0.285***	−0.404***	−0.398***	−0.386***	−0.315***
	(−3.56)	(−4.08)	(−3.25)	(−3.04)	(−4.26)	(−4.63)	(−3.37)	(−3.07)
Post × Treatment × TechProximity	0.319	0.182	1.028***	0.971***	0.709***	0.507**	1.235***	1.137***
	(1.34)	(0.89)	(3.19)	(3.47)	(2.76)	(2.24)	(3.57)	(3.72)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.858	0.854	0.847	0.852	0.855	0.852	0.844	0.849
Observations	25,981	37,528	16,822	24,084	20,837	30,246	14,689	21,188
Panel C. Industry entry costs: *Ln(PPE)*								
Post × Treatment	0.540***	0.495***	0.592***	0.750***	0.563***	0.525***	0.625***	0.801***
	(5.32)	(4.81)	(4.91)	(6.40)	(5.06)	(4.62)	(4.68)	(6.24)
Post × Treatment × Ln(PPE)	−0.071***	−0.057***	−0.100***	−0.110***	−0.072***	−0.061***	−0.099***	−0.112***
	(−3.98)	(−3.19)	(−4.60)	(−5.18)	(−3.58)	(−3.01)	(−4.14)	(−4.83)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.824	0.819	0.818	0.819	0.821	0.818	0.813	0.815
Observations	78,503	113,090	48,651	69,551	61,968	89,682	41,947	60,444
Panel D. Industry consumer switching costs: *TotalProdSimilarity*								
Post × Treatment	−0.178***	−0.167***	−0.072	−0.032	−0.146***	−0.143***	−0.054	−0.014
	(−5.09)	(−4.66)	(−1.61)	(−0.73)	(−3.55)	(−3.41)	(−1.05)	(−0.27)
Post × Treatment × TotalProdSimilarity	0.019**	0.025***	0.024**	0.023**	0.018**	0.023***	0.025***	0.024**
	(2.22)	(2.97)	(2.49)	(2.25)	(2.04)	(2.70)	(2.59)	(2.36)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.826	0.823	0.818	0.820	0.822	0.821	0.812	0.816
Observations	77,597	111,722	48,100	68,721	61,165	88,458	41,446	59,687

Notes. This table studies the impacts of industry characteristics on the relations between common institutional ownership and product market fluidity using DID models that exploit a set of mergers between financial institutions. Panels A–D study the incremental effect of industry concentration (*HHI*), technology proximity (*TechProximity*), entry costs (Ln(*PPE*)), and consumer switching costs (*TotalProdSimilarity*) on our baseline results reported in Table 4. The dependent variable is fluidity measured in percentage points. The main independent variable is the triple interaction term between *Post*, *Treatment*, and either *HHI*, *TechProximity*, Ln(*PPE*), or *TotalProdSimilarity*. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted), imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models using (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. For brevity, we only report the coefficients of interest (all other estimates are reported in Online Appendix Table 4). See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.

***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

Thus, the quiet life (efficiency) channel tends to dominate in industries with higher (lower) entry costs.

Finally, we consider the moderating effect of consumer switching costs. We use the firm-level measure of total product similarity developed by Hoberg and Phillips (2016) (*TotalProdSimilarity*) as a proxy for the degree of product differentiation and consumer switching costs (e.g., Hotelling 1929, Chamberlin 1933). Specifically, total product similarity is higher when a greater number of rivals offer more similar products to the firm's own product portfolio (based on their 10-K product description language). Total product similarity is inversely related to the switching costs for consumers because they can choose alternative vendors more easily in markets with low product differentiation.

The results from this analysis are presented in panel D of Table 5. Similar to prior tables, we only report our coefficient of interest associated with a triple-interaction term between *Post*, *Treatment*, and *TotalProdSimilarity* (other results are reported in Online Appendix Table 4). We find that the coefficients on the triple-interaction term are significantly positive at least at the 5% level in all the regressions. These results suggest that firms with higher common ownership face greater product market threats from rivals in industries with less differentiated products, that is, in industries with lower consumer switching costs. These findings are consistent with both the efficiency and quiet life channels (Hypotheses 5A and 5B). Finally, to provide additional perspective, in column (3), the coefficient on *Post* × *Treatment* is −0.072 and the coefficient on *Post* × *Treatment* × *TotalProdSimilarity* is 0.024 in column (3). This implies that the coefficient on *Post* × *Treatment* is positive (negative) in industries with *TotalProdSimilarity* greater (less) than 3.00 (which is less than the 75th percentile value of *TotalProdSimilarity*). Thus, the efficiency (quiet life) channel tends to dominate in industries with lower (higher) consumer switching costs.

Overall, our baseline and cross-sectional results on product market fluidity suggest that common ownership can promote greater product market dynamism, but this effect is heterogeneous across industries. Specifically, consistent with the efficiency (quiet life) channel, we find that common ownership can induce greater (lesser) product market competition in industries characterized by lower (higher) concentration, greater (smaller) technology proximity, lower (higher) entry costs, and lower (higher) consumer switching costs.

5. Additional Tests on New Product Development and Firm Investments

We next study changes in new product development and investment policies to gauge how cross-held firms alter their business strategies to arrive at more dynamic product markets.

5.1. Common Ownership and New Product Development

Studying the product development process can facilitate our understanding of whether and how firms benefit from industry knowledge spillover promoted under common ownership. Additionally, new product development in commonly owned firms can also explain why common owners simultaneously hold multiple firms in an industry, especially viewed in light of our finding that common ownership, on average, spurs great product market competition.

Following Hoberg and Phillips (2010), we measure a firm's new product development by studying change in the size of the firm's 10-K product description language, using the same DID regression models. Similar to our analysis on fluidity, we begin with a baseline DID model in which the dependent variable is now the number of words in 10-K product description sections. With merger × year and merger × firm fixed effects, our model identifies the within-merger firm change in the size of the firm's product description section after institutional mergers.

Panel A of Table 6 reports the results from our analysis of product development after an increase in common ownership. For brevity, we only report the coefficients of interest. The detailed results are reported in Online Appendix Table 5. Our baseline results suggest a weakly positive impact of common ownership on the change in 10-K product descriptions after institutional mergers. Specifically, the coefficients are significantly positive at least at the 10% level in three out of the eight regressions but are never significantly negative.

To understand the heterogeneous effect of common ownership, we also estimate a triple difference regression model by interacting *Post* × *Treatment* with the four industry characteristics analyzed in Section 4.4. Panel B of Table 6 shows that industry concentration generally has a significantly negative effect on the relation between common ownership and new product development as suggested by the coefficients on the triple interaction terms *Post* × *Treatment* × *HHI*. These coefficients are significantly negative at least at the 10% level in six out of the eight regressions. They are insignificantly different from zero in the remaining two regressions. Further, panel C of Table 6 shows a positive coefficient on the triple interaction term between *Post* × *Treatment* × *TechProximity* in all models, six of which are significant at least at the 10% level.

In the last two panels of Table 6, we analyze how the impact of common ownership on new product development varies with industry entry and consumer switching costs. We document that the positive effect of common ownership on product descriptions is weaker in industries with high entry costs in panel C. Specifically, the estimated coefficient is statistically significant in six out of eight regression models at least at the 10% level.

Table 6. The Relation Between Common Ownership and New Product Development

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
Panel A. Baseline								
Post × Treatment	0.051*	0.081**	−0.000	0.036	0.044	0.080**	−0.001	0.031
	(1.71)	(2.50)	(−0.00)	(0.87)	(1.27)	(2.14)	(−0.02)	(0.66)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.821	0.801	0.809	0.794	0.817	0.798	0.805	0.789
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel B. Cross-section – industry concentration: *HHI*								
Post × Treatment	0.153***	0.206***	0.109*	0.162***	0.133**	0.191***	0.098	0.153**
	(3.37)	(4.33)	(1.92)	(2.76)	(2.57)	(3.55)	(1.50)	(2.32)
Post × Treatment × HHI	−0.415***	−0.481***	−0.255*	−0.278*	−0.314*	−0.393**	−0.164	−0.236
	(−3.76)	(−4.25)	(−1.72)	(−1.84)	(−1.88)	(−2.37)	(−0.79)	(−1.13)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.821	0.801	0.809	0.794	0.817	0.798	0.805	0.790
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel C. Cross-section – industry technology proximity: *TechProximity*								
Post × Treatment	−0.210**	−0.193**	−0.331***	−0.263**	−0.268***	−0.265***	−0.383***	−0.314**
	(−2.53)	(−2.32)	(−3.12)	(−2.40)	(−2.81)	(−2.78)	(−3.23)	(−2.56)
Post × Treatment × TechProximity	0.275	0.318	0.550*	0.534*	0.529**	0.624**	0.773**	0.728**
	(1.24)	(1.41)	(1.83)	(1.71)	(2.03)	(2.31)	(2.31)	(2.06)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.795	0.776	0.790	0.777	0.793	0.775	0.787	0.774
Observations	25,981	37,528	16,822	24,084	20,837	30,246	14,689	21,188
Panel D. Cross-section – industry entry costs: *Ln(PPE)*								
Post × Treatment	0.340***	0.433***	0.158	0.277**	0.340***	0.445***	0.144	0.275**
	(3.78)	(4.52)	(1.40)	(2.26)	(3.26)	(4.00)	(1.12)	(1.99)
Post × Treatment × Ln(PPE)	−0.052***	−0.062***	−0.029	−0.042*	−0.054***	−0.064***	−0.026	−0.042*
	(−3.24)	(−3.68)	(−1.41)	(−1.94)	(−2.85)	(−3.25)	(−1.13)	(−1.72)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.822	0.803	0.810	0.795	0.818	0.800	0.806	0.790
Observations	78,503	113,090	48,651	69,551	61,968	89,682	41,947	60,444
Panel E. Cross-section – industry consumer switching costs: *TotalProdSimilarity*								
Post × Treatment	−0.092***	−0.079**	−0.073*	−0.047	−0.103***	−0.077**	−0.082*	−0.057
	(−3.00)	(−2.41)	(−1.75)	(−1.09)	(−2.80)	(−1.98)	(−1.68)	(−1.12)
Post × Treatment × TotalProdSimilarity	0.021***	0.024***	0.022**	0.023***	0.021***	0.024***	0.022**	0.024***
	(3.17)	(3.84)	(2.47)	(2.66)	(3.22)	(3.74)	(2.53)	(2.71)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.823	0.806	0.811	0.797	0.818	0.802	0.806	0.792
Observations	77,597	111,722	48,100	68,721	61,165	88,458	41,446	59,687

Notes. This table studies the relation between common institutional ownership and firms' product descriptions (number of words) in the annual report using DID models that exploit a set of mergers between financial institutions. Panel A reports the baseline regression results. Panels B–E study the incremental effect of industry concentration (*HHI*), technology proximity (*TechProximity*), entry costs (Ln(*PPE*)), and consumer switching costs (*TotalProdSimilarity*) on the baseline results reported in panel A. The dependent variable, *Product Description*, is the number of words in the firm's 10-K product descriptions (Hoberg and Phillips 2010). The main independent variable is the double or triple interaction term between *Post*, *Treatment*, and either *HHI*, *TechProximity*, Ln(*PPE*), or *TotalProdSimilarity*. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted) imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models using (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. For brevity, we only report the coefficients of interest (all other estimates are reported in Online Appendix Table 5). See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.

***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

Moreover, panel D of Table 6 shows that the coefficient on the triple interaction term *Post* × *Treatment* × *TotalProdSimilarity* is statistically significant at least at the 5% level in all regression models. These cross-sectional results suggest that positive common ownership shocks have a positive (negative) impact on new product development in industries that are less (more) concentrated and in which firms are technologically more (less) proximate, entry costs are lower (higher), and consumer switching costs are smaller (larger). Overall, when we view our product market fluidity and new product development results together, it appears that economic environments in which positive common ownership shocks promote greater (lesser) product market competition are also the same settings in which we observe greater (lesser) product expansion. In other words, we observe greater (lesser) product market dynamism and new product development in industry environments more conducive to spillover (market power) opportunities.

5.2. Common Ownership and Firm Investments

In this section, we examine the relation between firm investments and common ownership based on the predictions of the theoretical model in López and Vives (2019). Using our DID models, we study the overall impact of common ownership on investments as well as incremental impacts of industry concentration and technological proximity on the relation between common ownership and investments. We use the sum of capital expenditures, research and development (R&D) expenditures, and acquisition expenses as a proxy for investments (we obtain similar results excluding acquisition expenses). In our investment regression models, we control for several determinants of investments such as Tobin's Q, cash flows, firm size, leverage, and cash (see, e.g., Foucault and Frésard 2014, Asker et al. 2015).

Panel A of Table 7 shows our baseline DID estimates. For brevity, we only report the coefficients of interest. The detailed results are reported in Online Appendix Table 6. The coefficients on the interaction term *Post* × *Treatment* are positive in all models and statistically significant at least at the 1% level in four out of eight models. These results suggest that, on average, firms tend to increase investments when they share common ownership with competitors, which is consistent with the efficiency channel. Viewed in isolation, this baseline result is also consistent with the product differentiation channel. Recall that we find that, on average, average similarity goes up after a positive shock to common ownership. Taken together, this baseline investments result is only consistent with the efficiency channel.[11]

Regarding cross-sectional variations on the effect of common ownership on investments, we first find a negative and significant interactive effect of *HHI* at least at the 10% level in seven of eight regression models in panel B of Table 7. The coefficients on *Post* × *Treatment* × *HHI* in remaining model is also negative but slightly outside conventional levels of significance. This result is consistent with investment spillovers predicted in López and Vives (2019) as well as the idea that firms with higher common ownership reduce investments to enjoy a quiet life in a collusive equilibrium (Hicks 1935). This negative coefficient is, however, inconsistent with the product differentiation channel because it predicts greater (lesser) investments by firms subject to the common ownership shock in more (less) concentrated industries.

Finally, in panel C of Table 7, we test whether technology similarity affects the investment-promoting effect of common ownership as per López and Vives (2019). We generally find that the positive effect of common ownership on investments is stronger in industries with greater technology similarity. Specifically, the coefficient is statistically significant at the 10% (5%) level in six (five) out of eight regression models. Taken together with our previous findings on fluidity as well as new product development, these results provide evidence supporting the efficiency channel of common ownership through promoting investment spillovers between competitors.

5.3. Common Ownership and Operating Performance

Finally, using the same DID setting, we study the impact of common ownership on firm-level operating performance. We use operating ROA to proxy for operating performance, which we compute as earnings before depreciation and amortization divided by total assets. Following Giroud and Mueller (2010), we control for HHI, firm size, firm size square, and firm age. The results from this analysis are reported in Table 8. Panel A shows our baseline DID estimates. Again, for brevity, we only report the coefficients of interest. The detailed results are reported in Online Appendix Table 7. The coefficients on the interaction term *Post* × *Treatment* are positive in all models and statistically significant at least at the 10% level in six out of eight models. These baseline results suggest that, on average, firms tend to achieve better operating performance when they share greater common ownership with competitors. These results are consistent with all the proposed channels because an increase in common ownership may not only increase spillover benefits, but also the ability to collude.

Regarding cross-sectional variation arising from industry concentration on the effect of common ownership on operating performance, we find that the coefficients on *Post* × *Treatment* × *HHI* are significantly negative at least at the 5% level in seven of eight regression models in panel B of Table 8. These negative coefficients are consistent with efficiency as the "dominant" channel. Finally, in panel C of Table 8, we examine whether the impact of an exogenous positive shock on common ownership on operating performance is impacted by technology proximity. We find that the

Table 7. The Relation Between Common Ownership and Firm Investments

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
Panel A. Baseline								
Post × *Treatment*	0.006***	0.007***	0.002	0.003	0.007***	0.007***	0.003	0.005
	(2.62)	(2.99)	(0.51)	(1.21)	(2.87)	(3.05)	(0.84)	(1.56)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.522	0.504	0.501	0.487	0.509	0.490	0.495	0.480
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel B. Cross-section – industry concentration: *HHI*								
Post × *Treatment*	0.010***	0.007**	0.005	0.007*	0.012***	0.010***	0.005	0.007
	(2.86)	(2.25)	(1.25)	(1.69)	(3.27)	(2.85)	(1.19)	(1.63)
Post × *Treatment* × *HHI*	−0.024**	−0.015	−0.035***	−0.040***	−0.035***	−0.030**	−0.032*	−0.038**
	(−2.39)	(−1.59)	(−2.66)	(−3.26)	(−2.69)	(−2.51)	(−1.94)	(−2.47)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.522	0.504	0.501	0.487	0.509	0.491	0.495	0.480
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel C. Cross-section – industry technology proximity: *TechProximity*								
Post × *Treatment*	0.006	0.005	−0.005	−0.010	0.003	0.001	−0.010	−0.014
	(0.79)	(0.73)	(−0.55)	(−1.14)	(0.34)	(0.14)	(−0.97)	(−1.45)
Post × *Treatment* × *TechProximity*	0.025	0.016	0.052**	0.050**	0.040**	0.034*	0.066***	0.065***
	(1.37)	(0.98)	(2.35)	(2.53)	(2.00)	(1.83)	(2.78)	(2.94)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.514	0.498	0.491	0.482	0.496	0.481	0.477	0.470
Observations	25,981	37,528	16,822	24,084	20,837	30,246	14,689	21,188

Notes. This table studies the relation between common institutional ownership and corporate investments using DID models that exploit a set of mergers between financial institutions. Panel A reports the baseline regression results. Panels B and C study the incremental effect of industry concentration (*HHI*) and technology proximity (*TechProximity*) on investments. The dependent variable is the sum of capital expenditures, R&D expenditures, and acquisition expenses scaled by total assets at the beginning of the year. The main independent variable is the double or triple interaction term between *Post*, *Treatment*, and either *HHI* or *TechProximity*. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted), imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models using (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. For brevity, we only report the coefficients of interest (all other estimates are reported in Online Appendix Table 6). See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.
***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

coefficients on *Post* × *Treatment* × *TechProximity* are significantly positive at the 1% level in all eight regression models. These cross-sectional results suggest that operating performance is higher in environments in which spillover opportunities are greater. When these results are jointly evaluated, the efficiency channel appears to be dominant in explaining the superior operating performance following an exogenous increase in common ownership.[12]

6. Conclusion

The existing empirical evidence on the competitive effects of common ownership remains inconclusive. We offer new evidence on this topic by studying the relation between common ownership and competitive threats in product spaces rather than prices or outputs. We use fluidity, developed by Hoberg et al. (2014), as a measure of the product market threats faced by a firm. To establish causality, we estimate a DID model using mergers between financial institutions as an exogenous positive shock to common ownership. Our baseline results suggest that higher common ownership leads to greater fluidity, higher average product similarity, more product development, and higher investments. These findings suggest that, on average, common ownership spurs dynamism in product spaces rather than tacit collusion.

Table 8. The Relation Between Common Ownership and Operating Performance

Holding periods	Any Quarters				4+ Quarters (Predicted)			
Merger events	All		Recommended		All		Recommended	
Event window	(−2, +2) (1)	(−3, +3) (2)	(−2, +2) (3)	(−3, +3) (4)	(−2, +2) (5)	(−3, +3) (6)	(−2, +2) (7)	(−3, +3) (8)
Panel A. Baseline								
Post × *Treatment*	0.006***	0.006***	0.003	0.003	0.007***	0.007***	0.005**	0.005**
	(3.49)	(3.41)	(1.35)	(1.50)	(3.64)	(3.66)	(2.07)	(2.17)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.741	0.720	0.730	0.701	0.730	0.707	0.726	0.697
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel B. Cross-section – industry concentration: *HHI*								
Post × *Treatment*	0.010***	0.011***	0.007**	0.007**	0.010***	0.011***	0.008**	0.009***
	(4.01)	(4.08)	(2.25)	(2.46)	(3.48)	(3.86)	(2.42)	(2.80)
Post × *Treatment* × *HHI*	−0.024***	−0.023***	−0.028***	−0.027***	−0.013	−0.018**	−0.025***	−0.029***
	(−3.59)	(−3.64)	(−3.57)	(−3.55)	(−1.59)	(−2.23)	(−2.64)	(−3.00)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.742	0.720	0.730	0.702	0.730	0.707	0.726	0.697
Observations	78,594	113,225	48,678	69,589	62,024	89,759	41,969	60,474
Panel C. Cross-section – industry technology proximity: *TechProximity*								
Post × *Treatment*	0.000	−0.001	−0.014**	−0.013**	0.001	0.000	−0.008	−0.009
	(0.06)	(−0.21)	(−2.07)	(−1.98)	(0.24)	(0.08)	(−1.09)	(−1.26)
Post × *Treatment* × *TechProximity*	0.041***	0.047***	0.055***	0.055***	0.045***	0.049***	0.048***	0.055***
	(3.02)	(3.34)	(3.43)	(3.37)	(3.04)	(3.21)	(2.72)	(2.98)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Merger-firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.767	0.752	0.758	0.736	0.756	0.738	0.754	0.729
Observations	25,981	37,528	16,822	24,084	20,837	30,246	14,689	21,188

Notes. This table studies the relation between common institutional ownership and firm operating performance using DID models that exploit a set of mergers between financial institutions. Panel A reports the baseline regression results. Panels B and C study the incremental effect of industry concentration (*HHI*) and technology proximity (*TechProximity*) on operating performance. The dependent variable is operating ROA computed as operating income before depreciation and amortization divided by total assets. The main independent variable is the double or triple interaction term between *Post*, *Treatment*, and either *HHI* or *TechProximity*. We consider two sets of treatment groups: (i) Any Quarters, including all firms that become commonly owned with their rivals after institutional mergers, and (ii) 4+ Quarters (Predicted) imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out 12 quarters after the mergers. Any treated firms included in (i) but not in (ii) are dropped from the regression models using (ii). There are two sets of merger events: All mergers, defined as the union of merger events provided by He and Huang (2017) and Lewellen and Lowry (2021), and Recommended merger events that occurred outside the 2007–2008 global financial crisis. For brevity, we only report the coefficients of interest (all other estimates are reported in Online Appendix Table 7). See Appendix A for the complete list of variable definitions. Standard errors are clustered at the firm-level. *t*-statistics are in parentheses.

***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

Our cross-sectional results suggest that the positive impact of common ownership on product market dynamism is driven by environments in which it is easier to take advantage of knowledge spillovers; that is, in industries characterized by low concentration and high technology similarity. However, common ownership can hinder product market dynamism in industries more prone to quasi-monopoly outcomes in product spaces, that is, in industries that are highly concentrated, which have high entry costs, or in which consumer switching costs are high. These latter results provide support for the quiet life channel. Our cross-sectional examination of firms' new product development and investments corroborates these conclusions.

This paper makes the following contributions. First, we directly contribute to the recent literature on the competitive effects of common ownership by showing that it can lead to greater competitive threats and more dynamic product spaces. Second, we contribute to the recent debate on the competitive effects of common ownership. Our findings suggest that caution needs to be exercised in proposing and implementing any additional regulatory requirements on the ownership of competing firms by financial institutions because they can potentially be welfare-destroying in industries with strong spillover opportunities (López and Vives 2019). In this regard, we believe our paper provides insights that can inform the policy decisions of global regulatory agencies that are concerned about the secular increase in common ownership and its potential detrimental effects on product market competition and firms' strategic decisions.

Finally, we conclude by noting that common ownership can also influence firms beyond product-market measures. For example, financial institutions may share common values concerning climate exposures and may commonly own shares to influence climate investments (Akey and Appel 2019, Krueger et al. 2020, and Dimson et al. 2023). Therefore, common ownership of firms by institutional investors can lead to coordinated efforts to achieve shared social objectives. Thus, our focus on studying the competitive effects of common ownership is by no means the last word on common ownership's potential for broad societal impact through, for example, its potential impact on environmental and social issues. We believe that future research on common ownership along these lines will be fruitful.

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Appendix A. Definition of Variables

Table A.1 provides the complete list of definition of variables used in this paper. Compustat and CRSP mnemonics are reported in parenthesis with bold fonts, for example, Total assets (**at**).

Table A.1. Definition of Variables

Variables	Description
Fluidity and TNIC-level variables	
Fluidity	Product market fluidity by Hoberg et al. (2014).
AvgSimilarity	The average product similarity measure by Hoberg and Phillips (2016) between a firm and its product market competitors identified using TNIC.
Product Description	Number of words in the firm's 10-K product description language by Hoberg and Phillips (2010).
HHI	Herfindahl–Hirschman index of sales. We use the TNIC-level HHI available at the Hoberg–Phillips Data Library.
TechProximity	The extent to which a firm's technology is similar to the technology of its product market competitors. We use the Jaffe (1986) measure of cosine similarity (ranging between zero and one) between the patent portfolios of two firms, in which a firm's patent portfolio includes all patents filed in a three-year window from $t-2$ to t (Bena and Li 2014). The firm's patent portfolio is a vector of the shares of the firm's patents in 456 patent classes according to the U.S. Patent and Trademark Office classifications. We compute the median of technology proximity between a firm and its TNIC competitors. For the calculation of firm-level patent portfolios, we use Duke Innovation & Scientific Enterprises Research Network patent data sets made by Arora et al. (2021).
PPE	TNIC-level median value of gross property, plant, and equipment (**ppegt**) in 2012 dollars using the U.S. gross domestic product (GDP) deflator.
TotalProdSimilarity	The extent to which a firm's product description is similar to its product market competitors' product descriptions and how many competitors exist in the product market. We compute the total product similarity between a firm and its TNIC competitors as the sum of pair-level product similarity measures by Hoberg and Phillips (2016).
Common ownership	
MHHID	O'Brien and Salop (2000): $$\underbrace{\sum_j \sum_k s_j s_k \frac{\sum_i \gamma_{ij} \nu_{ik}}{\sum_i \gamma_{ij} \nu_{ij}}}_{MHHI} = \underbrace{\sum_j s_j^2}_{HHI} + \underbrace{\sum_j \sum_{k \neq j} s_j s_k \frac{\sum_i \gamma_{ij} \nu_{ik}}{\sum_i \gamma_{ij} \nu_{ij}}}_{MHHID},$$ where s_j is the market share of firm j, ν_{ij} is the ownership share of firm j accruing to investor I, γ_{ij} is the control share of firm j exercised by investor I, and k indexes firm j's competitors.

Table A.1. (Continued)

Variables	Description
Cross	Hansen and Lott (1996): $$Cross_j = \sum_k \left(\sum_i \alpha_{ij} \times \sum_i \alpha_{ik} \right),$$ where k indexes the competitors of firm j and i indexes the investors holding common ownerships of firm j and k at the same time.
GGL	Gilje et al. (2020): $$GGL_j = \sum_{k \neq j} \sum_i \alpha_{ij} \times g(\beta_{ij}) \times \alpha_{ik},$$ where k indexes the competitors of firm j and i indexes the investors holding common ownerships of firm j and k at the same time. $g(\beta_{ij})$ represents investor i's incentives to be attentive to firm j's operation as a function of the proportion of firm j's stock values in the investor's portfolio (β_{ij}). We follow the baseline specification of Gilje et al. (2020) that assumes a linear and positive relation between investors' attention and their ownership level: $g(\beta_{ij}) = \beta_{ij}$.
Firm characteristics	
Assets	Total assets (**at**) in 2012 dollars with U.S. GDP deflator.
Age	One plus the number of fiscal years in Compustat.
Tobin's Q	(Total assets (**at**) + (Common share price (**prcc_f**) × Common shares outstanding (**csho**)) − Common equity (**ceq**) − Deferred taxes balance sheet (**txdb**))/Total assets (**at**).
ROA	Operating income before depreciation and amortization (**oibdp** or equivalently **sale** − **cogs** − **xsga**)/Total assets (**at**).
Leverage	Short-term (**dlc**) and long-term debt (**dltt**)/Total assets (**at**).
Cash	Cash (**che**)/Total assets (**at**).
Investments	Sum of capital expenditures (**capx**/**at**), R&D expenditures (**xrd/at**), and acquisition expenses (**aqc**/**at**). We set **xrd** and **aqc** to zero for missing values.
Return volatility	Standard deviation of daily stock return volatility for a year.

Appendix B. Persistence of Common Ownership After Institutional Mergers

In our DID regression model, we identify two sets of treatment firms: (i) Any Quarters, including all firms that become commonly owned with their rivals after a merger between financial institutions, and (ii) 4+ Quarters (Predicted) imposing an additional condition to (i) that such common ownership is anticipated to last at least 4 out of 12 quarters after the merger. In this section, we first describe the theoretical underpinnings behind exploiting mergers between financial institutions as shocks to common ownership. We then use this framework to motivate the choice of explanatory variables in our model to predict treated firms for which common ownership is likely to persist. Finally, we provide details of our prediction model for the persistence of common ownership used to determine the treatment group for (ii). Notably, the control firms remain the same in (i) and (ii).

The basic premise behind our paper as well as other common ownership studies is that, on average, an increase in common ownership increases the value of the commonly held firms either because of the greater benefits they derive from collusion (e.g., Azar et al. 2018, 2022a) or through higher knowledge spillovers that lead to more dynamic product markets through improved products, new products, new markets, etc. (e.g., He and Huang 2017, López and Vives 2019). We are likely to observe changes in product market outcomes when an exogenous shock alters the equilibrium configuration of common ownership levels.

Mergers between financial institutions can be between commercial banks, investment banks, securities firms, insurance companies, and asset management firms, etc. These mergers serve as exogenous shocks in our setting because they are motivated either by deregulation and/or business strategy considerations and, consequently, are unrelated to product market strategies of individual firms (He and Huang 2017). When the asset management businesses of the merging institutions are combined, funds with the same style may be merged, new funds introduced, and some old funds discontinued or sold. Since the assets under management (AUM) for the merged firm are larger, the restructuring likely results in a decline in active shares in the overall portfolio because of the price impact of trading larger portfolios. As such, to maximize dollar returns, the merged institution tends to maintain a limited portion of active investments, becoming more passive in most other investments (Berk and Green 2004, Cremers and Petajisto 2009). In other words, the costs that arise from a lack of diversification go up after the merger because of an increase in scale. As a result, we are likely to observe a reduction in the weights of treated firms and their industries in the combined portfolio that are overweighted relative to a passive benchmark at the time of the merger.

In addition, because of increased concentration resulting from higher common ownership, the costs of tacit collusion also increase because of greater scrutiny from antitrust authorities. Thus, ceteris paribus, merged institutions that face higher diversification costs and costs of tacit collusion from their investments in treated firms reduce their investments in these firms and, as a result, forgo some of the benefits of common ownership. Consequently, common ownership is less rigid for these firms. Conversely, if diversification costs and costs of tacit collusion are lower, then common ownership in treated firms is likely to persist, and the common owners can take advantage of the benefits of common ownership. In sum, common ownership may not fully revert back to premerger levels because the effects of the reconfigured costs are heterogeneous across firms in the combined portfolio of the merging institutions, thereby resulting in a new equilibrium.

Our goal is to predict the probability of retaining common ownership after institutional mergers. Therefore, the sample in the prediction model includes only the treatment group of firms as defined in our DID tests. We estimate the following probit regression models:

$$\begin{aligned} Treat4qtr_{i,j,t} = {} & \beta_0 + \beta_1 FirmHolding\_AUM_{i,j,t-1} \\ & + \beta_2 IndHolding\_AUM_{i,j,t-1} + \beta_3 HHI_{i,j,t-1} \\ & + \beta_4 Ln(MktCap)_{i,j,t-1} + \beta_5 PctHolding_{i,j,t-1} \\ & + \delta_j + \epsilon_{i,j,t}, \end{aligned} \tag{B.1}$$

where i, j, and t indicate each firm, merger, and time period in our data set. *Treat4qtr* is an indicator variable that equals one if the merger-induced common ownership lasts for at least 4 out of 12 quarters after the merger and zero otherwise.

We construct two variables to capture the diversification needs of the merging institution. *FirmHolding_AUM* is calculated as the sum of each merging institution's dollar value holding of firm i divided by the sum of the AUM, net of a benchmark weight that equals firm i's market capitalization as a fraction of the total market capitalization of all firms included in the TNIC data set. Similarly, *IndHolding_AUM* is calculated as the sum of each merging institution's dollar value holding of industry j to which firm i belongs divided by the sum of the AUM, net of a benchmark weight that equals industry j's market capitalization as a fraction of the total market capitalization of all firms included in the TNIC data set. A higher value of these variables indicates that the merger induces an overrepresentation of the firm or industry in the merged institution's portfolio relative to a passive index, thereby imposing costs because of suboptimal diversification. We, therefore, predict a negative relation between persistence of common ownership with these two variables.

We include *HHI* in the regression model because higher common ownership is more likely to bring the attention of antitrust authorities in more concentrated industries and, thus, increase the costs of tacit collusion. We, therefore, predict a negative relation between the persistence of common ownership with *HHI*. We also control for the market value of the firm (*MktCap*) and the percentage ownership of firm i by the merged institution (*PctHolding*). Finally, we include merger indicators (δ_j) in the regression model to improve the precision of the estimates. Our main findings remain similar after excluding merger indicators or fewer independent variables for a more parsimonious prediction model.

Our empirical prediction for the persistence of common ownership is obtained as follows. Suppose that, for a specific merger event in our sample period, say merger 1, there are 12 mergers that occurred in the previous 10 years of the announcement of merger 1. Using the same criteria used in our DID tests, we first identify treated firms for each of these 12 mergers. Among these treated firms, we determine which firms remain commonly held with their TNIC rivals for at least 4 out of 12 quarters after the merger, which determines the value of our dependent variable *Treat4qtr*. Then, we estimate the probit regression model and apply the parameters to merger 1 to calculate the probability of retaining common ownership for at least 4 out of 12 quarters after merger 1. Finally, we assign firms into the treatment group for 4+ Quarters (Predicted) if the estimated probability is greater than or equal to 50%. We repeat this procedure for each of 27 merger events used in our DID tests. It is worth noting that, by definition, the treatment group for 4+ Quarters (Predicted) is a subset of the treatment group for the Any Quarters sample. As stated earlier, the control group does not change for any given merger event; specifically, it is the same control group that we use in the Any Quarters sample.

We cannot perform such out-of-sample estimation for the mergers in the earlier years of our sample period because of the absence of prior merger events to estimate the model parameters. To overcome this constraint, for these prediction models, we combine our data set with the list of institutional mergers from Lewellen and Lowry (2021) that occur prior to the start of our sample period. This extended institutional merger sample allows us to perform out-of-sample predictions for the persistence of common ownership for each of the merger events in our DID tests. Specifically, we accomplish this goal by estimating separate prediction models using data on institutional mergers that occur in the 10 years prior to each merger event in our sample.

To illustrate the key results from our estimation briefly, we present the coefficients and marginal effects estimated from a pooled set of merger events (rather than reporting the results from each individual prediction model for a merger event using data on all the merger events in the previous 10 years of that specific merger event) in Table B.1.

In this pooled prediction model, consistent with diversification concerns, we find that a firm is less likely to be commonly held for at least 4 out of 12 quarters after the merger

Table B.1. Persistence of Common Ownership After Institutional Mergers

Dependent Variable: *Treat4qtr*	Coefficient	Marginal effect
FirmHolding_AUM	−0.263***	−0.082***
	(−2.69)	(−2.70)
IndHolding_AUM	−0.021***	−0.007***
	(−4.88)	(−4.89)
HHI	−0.879***	−0.272***
	(−13.38)	(−13.56)
Ln(MktCap)	0.136***	0.042***
	(20.29)	(20.97)
PctHolding	0.067***	0.021***
	(12.47)	(12.61)
Merger Dummies	Yes	
Pseudo R^2	0.209	
Observations	18,279	

***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively.

if the firm (industry) is more overweighted in the combined portfolio of the merging institutions relative to the value-weighted passive index portfolio. In addition, consistent with antitrust concerns, a firm is less likely to be commonly held for at least 4 out of 12 quarters after the merger if its industry is more concentrated. We additionally find that larger firms are more likely to be commonly held for at least 4 out of 12 quarters after the merger, which is consistent with the idea that institutional investments in more important firms tend to persist for longer periods of time. Finally, a firm is more likely to be commonly held for a longer period of time if the merged institution holds a higher fraction of its shares. This is likely to be a mechanical result driven by the fact that, at high levels of fractional ownership, the firm is still likely to qualify as being commonly held even if the merged institution were to divest some of its holdings in the firm.

In the pooled sample of treatment firms with the 0.5% cutoff of common ownership, the actual likelihood of retaining common ownership after the merger is 45.90%, whereas the predicted likelihood is 42.95%. In a similarly constructed sample of treatment firms with the 1% cutoff of common ownership, the actual likelihood of retaining common ownership after the merger is 34.48%, whereas the predicted likelihood is 22.42%. The concordance between the predicted 4+ Quarters treatment firms and the actual treatment firms is 70.47% (75.27%) in the 0.5% (1%) cutoff sample. The accuracy of the model and estimated parameters remains similar in the prediction models estimated on a 10-year rolling window basis.

Endnotes

[1] See, for example, Rotemberg (1984), Reynolds and Snapp (1986), Bresnahan and Salop (1986), Hansen and Lott (1996), O'Brien and Salop (2000), and López and Vives (2019) for theoretical arguments along these lines. Azar et al. (2018, 2022a,b) find empirical support for the market power effect in the airline and banking industries.

[2] A caveat is in order here. Increased fluidity may not always capture greater product market threats, for example, when firms actively differentiate their products to not compete in each other's product spaces.

[3] Our inferences remain unchanged if we drop these observations from the calculation or classify them into the control group because, among all the firm pairs held by merging institutions, only less than 0.1% of the pairs are already cross-held by both merging institutions.

[4] Our DID results are robust to the choice of identifying treatment and control firms based on a 1% cutoff. Thus, the choice of this cutoff does not alter the overall conclusions we reach in the paper. We report the results of these analyses in Online Appendix Table 2. We also provide a detailed discussion on the choice of common ownership cutoff in Online Appendix B.

[5] It is possible to identify the long-term treatment group based on whether the common ownership actually lasts (rather than is anticipated to last) four or more quarters after the merger. For example, for the 0.5% common ownership cutoff, we find that 60.1%, 54.8%, 50.2%, and 46.0% of the treated firms remain commonly owned in the one, two, three, and four quarters after the merger, respectively. However, such classification relies on an ex post sorting on outcomes, which creates an issue of simultaneity in which the decision to hold common ownership and the realized firm- or industry-level outcomes could be correlated. Our prediction-based treatment is not subject to this issue although it includes some misclassified firms in the treatment group as is the nature of any prediction model. Nevertheless, our findings remain very similar when we identify the long-term treatment group based on the actual length of common ownership after institutional mergers. These results are reported in Online Appendix Table 3.

[6] This test requires an additional data set of merger events that extends to the period before the first merger event in our sample. See Appendix B for details of the data set construction for our prediction models.

[7] We find that including these firms in the control group generates similar results.

[8] The sample size increases from the (−2, +2) to (−3, +3) window as our unit of observations is constructed at the firm–year level rather than aggregating pre- and post-event observations into a single observation. We include merger × year fixed effects to control for heterogeneity across each merger–year pair.

[9] Online Appendix Table 1 reports the descriptive statistics for the largest sample used in our DID model: All mergers with a (−3, +3) event window.

[10] Given that the drug (steel) industry is characterized by low (high) industry concentration, low (high) entry costs, and low (high) consumer switching costs, we expect a negative (positive) coefficient on *Post* × *Treatment* for the drug (steel) industry under the product differentiation channel. Our findings for these industries are the opposite of what is predicted by this channel.

[11] We conduct additional DID tests focused on R&D activities. Since R&D is an imprecisely measured variable, we construct a dummy variable that equals one if R&D expenditures are greater than zero and use it as the dependent variable. We find weak evidence that treatment firms are more likely to engage in R&D activities after financial institution mergers, which is consistent with our baseline investment results.

[12] We also examine announcement period abnormal returns (*CAR*) for treatment and control firms, which we estimate using the market model over the event window (−1, +1) days around institutional merger announcements. We find that the average *CAR* is 0.63% for treatment firms, 0.39% for control firms, and the difference is significant at 1%. In addition, we conduct cross-sectional tests with *CAR* as the dependent variable. The results also generally suggest that efficiency is the dominant channel and are reported in Online Appendix Table 8.

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